



A guide to design uncertainty-aware self-adaptive components in Cyber–Physical Systems

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ABSTRACT

Cyber–physical systems (CPS) need to be designed to deal with various forms of uncertainties associated with data contributing to the system's knowledge and its belief of the environment. Dealing with uncertainties requires adopting appropriate methods, which then allow for making the right decisions and carrying out the right actions based on imperfect information. However, CPS design is difficult and requires attention to many variables during the design process. For instance, choosing and incorporating methods for dealing with uncertainties into CPS design requires identifying the kinds of uncertainties at hand as well as knowledge of suitable methods and their application. While inspiration can be found in other CPS designs, the details of dealing with uncertainties in other CPSs can be confounded by domain-specific terminology, context, and requirements. To make this aspect of CPS design less daunting, we aim at providing an overview of approaches dealing with uncertainties in the design of CPS targeting collective behavior. To this end, we present a systematic review of relevant scientific projects with industrial leadership and synthesis of relations between system features, the kinds of uncertainties, and methods used to deal with them. The results provide an overview of uncertainties and challenges across different domains, and reason about a guide for designing uncertainty-aware self-adaptive components in CPS.

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1. Introduction

Computer systems help humans in performing tasks and overcoming real-life challenges. The complexity in designing each system varies depending on the domain [1] and the requirements (i.e., system properties) to be satisfied. A Cyber–Physical System (CPS) is an example of a complex system. It involves communication between hardware parts that realize the interaction with the physical world and the software parts that perform the necessary computations. Due to the advent of Industry 4.0 and the growth of using Internet-of-Things (IoT) [2], there is a need for cooperative behavior [3] between different devices leading to the emergence of Smart CPS (sCPS). The system parts are subject to changes in the environment that influence the satisfaction of system requirements. Therefore, the systems need to adapt to the changing environment to accomplish the desired goal. However, sCPS have many inherent uncertainties, which complicate adaptation decisions at runtime.

To design sCPS, first, we have to learn more about existing systems, how they address the uncertainties and what impacts the adaptation decisions they perform. The CPSs exist in different

domains, and their specification can use domain-specific terminology, context, and requirements. However, the developer can still benefit from investigating the design options in CPSs in other domains, involved uncertainties, and the methods to handle them before performing runtime adaptation.

To form a reusable knowledge foundation, we aim at performing a systematic review [4] that provides us with information about sCPS design and uncertainties from industrial demonstrators where we require three essential aspects to be covered: Self-adaptation, Cooperation, and Physical-World Interaction.

This study aims to describe and compare the existing approaches to handle uncertainties and how they impact the adaptation decisions reflecting the system requirements. The knowledge acquired helps learn more about possible relations between high-level and low-level aspects in the CPS design, involving uncertainties, aiming at enhancing and simplifying the design process.

The rest of the article is structured as follows: Section 2 discusses the state-of-the-art, Section 3 presents the method used for conducting the study. Section 4 describes the planning phase of the study, while Section 5 describes the conducting phase. The results are divided into two parts: Section 6 is based on the results of the systematic review of EU projects [5] and provides a classification of domains and challenges, along with the adaptation and uncertainties information. In Section 7, we

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reason about the results and model the relations between CPSs, adaptation, cooperation, and uncertainties. Based on the findings, we provide a more detailed view of methods and adaptation decisions as a base for a guide to design uncertainty-aware self-adaptive components in CPSs. Then, we demonstrate the use of one of the applications of the design guide on a representative demonstrator from the projects and conclude with a discussion on threats to validity in Section 8. Summary and discussion in Section 9 conclude the article.

2. Related work

Looking into the current related work, we go through state of the art about uncertainties in each of the aspects we target in the article, namely Self-Adaptation, Cooperation, and Physical-World Interaction.

Self-adaptation. A method to capture uncertainties in adaptation is presented in [6]. The work provides an approach, called Adapt Cases, that proposes an explicit description of the adaptability in use cases in *logical design*, which links the requirements and the technical design. The solution presents a meta-model structured into three packages for mapping the MAPE-K model, which are monitor (i.e., monitor and analyze), adaptation (i.e., plan and execute), and adaptation context (i.e., knowledge). Even though the work aims at linking the requirements and technical design through use cases, it is similar to our approach concerning employing the MAPE-K model to capture the impact of uncertainties on adaptation decisions that satisfy requirements. However, unlike our work, the approach does not go deeper into studying uncertainties or methods to handle them.

The authors in [7] aim at reducing the impact of uncertainties in quality evaluation. They define a taxonomy for uncertainties and study their sources. The study shows that multiple uncertainties could impact model evaluation. In [8], the study aims at defining a taxonomy of types of uncertainties, templates for their sources, occurrence, and their effect on requirements, design, and runtime levels. Even though the studies have multiple shared aspects with our study, they do not propose a well-defined development process for self-adaptive CPSs.

A survey by Villegas et al. [9] deals with mapping system properties in self-adaptive systems to Quality Attributes (QA). Its goal is to evaluate a self-adaptive system according to an adaptation metric derived from related quality attributes. This solution helps avoid the use of properties that could be unrelated to the adaptation goal (e.g., performance attribute). The work of Malavolta et al. [10] aims at identifying the application domains, challenges, goals, and solutions for architecting CPSs. Their study infers links between challenges and suggests solutions for architecting CPS.

The work presented in [11] proposes an approach, similar to ours, to develop self-adaptive systems (SAS). It starts by modeling and specifying uncertainties, generating reasoning rules, designing adaptation mechanisms, and decision-making and learning knowledge. The schemes to adapt and evolve are based on fuzzy logic and presented through four schemes: forward, backward, parameter-identification, and system-identification scheme. Nevertheless, the approach does not consider the specification of different uncertainties and the methods to handle them. Also, the main targeted output is to learn or optimize and adapt accordingly.

As for matching involvement of taxonomy for uncertainties in adaptation, in [12], the authors define a taxonomy for uncertainties, classify the types of uncertainties and match them to MAPE-K stages. However, they do not consider the properties and architectural aspects.

Cooperation. The authors in [13] perform a systematic review that targets Collaborative Self-Adaptive Systems (CSAS). The paper aims to help the developers select suitable learning models and techniques in CSAS based on a 3D framework for designing learning-enabled CSAC that determines autonomy, knowledge access, and behavior as key characteristics. The approach explains the impact of the key characteristics on the selection of the learning models and techniques used in the CSAS agents. This study intersects with ours regarding autonomy, knowledge, and behavior access in designing CPS. However, it does not explicitly address the uncertainties sources; rather, it focuses on learning methods used to handle the uncertain environment in general. The study is limited to learning methods and states; however, it leaves the relation between learning methods and uncertainties types as an open challenge. Musil et al. [14] aim at finding out application domains and types of uncertainties in self-adaptive CPSs, but not including properties. The study shows interest in connecting the properties of the system with architectural choices but does not capture the details of uncertainties on behavior.

Physical-world interaction. For embedded systems, the paper [15] discusses identifying kinds and sources of uncertainties and presents a set of techniques to mitigate the uncertainties. The techniques include feedback loops, making uncertainties explicit, apply robust design, and use *wide sampling*. The work intersects with ours in including sources and types of uncertainties and methods to handle them. However, it does not discuss the relations between the types of uncertainties and system properties. We also go a step further by introducing a taxonomy for the methods used to handle uncertainties.

3. Systematic review approach

To conduct the study, we aim at studying industrial demonstrators in ICT EU projects. We target ICT EU projects because they are led by industry and reflect the current trend of both industrial and academic research, and present demonstrators covering multiple design aspects.

More specifically, we study a set of industrial sCPS demonstrators in EU projects to learn what methods are used to handle uncertainties in the industry and how to identify uncertainties during collective sCPS design. To focus on relevant projects, we require three basic aspects of sCPS that the projects address:

1. **Self-adaptation.** This aspect represents the ability of a system to perform self-adaption and to make decisions depending on the context (sometimes called *smart*). It is a match to using the feedback loop to control the behavior.
2. **Cooperation.** This aspect represents communication among entities and coordination of behavior in a distributed fashion (i.e., collective behavior).
3. **Physical-World Interaction.** This aspect ensures that the system has a physical part, which means it has sensing (and usually also actuating) capabilities. This includes systems such as Embedded systems, Internet-of-Things, and Cyber-Physical Systems.

In this study, we follow the guidelines for performing systematic literature reviews (SLR) in software engineering presented by Kitchenham [4]. A systematic review is split into the planning, conducting, and reporting phases. The planning phase includes a definition of the research questions and clarifies the source of primary studies. It also determines inclusion and exclusion criteria, defines study procedures, a quality assessment method, and data extraction forms to extract data relevant to the research questions. The planning phase results in a protocol that has to be

approved by the researchers. In the conducting phase, we collect the data based on the extraction forms; based on them, we aim at answering the research questions. Finally, the reporting phase provides a detailed description of the results.

The SLR guidelines suggest that the primary studies be peer-reviewed papers. However, peer-reviewed papers do not guarantee the existence of real industry-relevant demonstrators, nor do they provide a comprehensible view of the demonstrators and their development challenges. Choosing the primary studies was discussed by Kitchenham [16]; the findings there show that the types of research questions in the study determine the selection of primary studies. Therefore, we chose deliverables of projects in place of peer-reviewed papers as primary studies.

Based on the deliverables, we access both the results of the publications and the description of the use cases. Also, we ensure to count each relevant use case just once in the domain.

The study (schematically shown in Fig. 1) started with identifying the research questions, selection criteria and selection procedures, study quality assessment, data extraction forms, and data synthesis. The protocol was validated by internal and external researchers [17]. After the conducting phase, in the reporting phase, we present the extracted data based on several existing classifications and update or propose new classifications that are important for designing uncertainty-aware self-adaptive components in CPS.

4. Planning the study

In this section, we present steps of the planning phase (i.e., we updated the protocol slightly after the validation.) of the systematic review, which are: research questions, search for primary studies, selection criteria, selection procedures, quality assessment, data extraction, and data synthesis.

4.1. Research questions

Our study is driven by the following research questions:

RQ1: What are the typical domains for collective sCPS?

The domains reflect the variety of architectures in the projects. Including various domains provides the developers with a good foundation to get oriented among the current trends in the sCPS application development and identifying differences in their interests.

RQ2: What are the challenges addressed by collective sCPS?

We define the challenges as the problems that the system should resolve to provide the expected services. Thus, it is necessary to understand the relation between the challenges and extra-functional system properties that services should satisfy using collective behavior.

RQ3: How do uncertainties relate to runtime adaptation decision in collective sCPS?

Addressing some challenges leads to the need for performing runtime adaptation decisions, which are impacted by uncertainties. In this light, the developer needs to understand the sources of uncertainties that emerge from CPS parts, the types of uncertainties to model, the methods to handle these uncertainties, and their relation to the self-adaptation decisions and cooperation between components to provide the expected services or to reach the desired goal.

4.2. Search for primary studies

In the context of this study, the primary studies are represented by European projects with industrial leadership listed in the CORDIS database at the European Commission website (<https://cordis.europa.eu/>). Specifically, we target the projects under the “Information and Communication Technologies (ICT)” calls. These are high-quality projects reporting working (physical) demonstrators in their deliverables. When searching for a preliminary sample of related projects, the results showed that most projects returned by the query were accepted under ICT calls from the EU Commission, confirming our selection for the calls. The CORDIS database can be considered trustworthy and provides results that are independent of commercial search engines.

We considered the projects from the past ten years to provide a representative sample of recent results in this research area. Specifically, we targeted the FP7-ICT and Horizon2020-ICT calls [18]. The FP7 calls span from 2007 to 2013, while the H2020 calls span from 2014 to 2016. We did not include H2020 projects that started in 2016 and later because these only have preliminary results.

4.3. Study selection criteria

We defined four basic selection criteria, shown in Table 1. Selection criteria 1, 2, and 3 contain a set of representative keywords, which we require to be included in the call and project descriptions. These keywords were extracted manually from a sample of related calls and project descriptions, and then filtered to obtain a more representative set.

We include projects that satisfy *all* of the following inclusion criteria (IC):

Calls

IC1 The call was made in 2007–2016.

IC2 The title of a call contains at least one keyword from each selection criteria 1, 2, and 3 (c.f. Table 1).

IC3 The description of a call contains at least one keyword from each of the selection criteria 1, 2, and 3.

Projects

IC4 The project title contains at least one keyword from any of the selection criteria 1, 2, and 3.

IC5 The project description contains at least one keyword from each of the selection criteria 1, 2, and 3.

IC6 The project results contain deliverables.

Use cases

IC7 Challenges in the use case target collective behavior.

IC8 The use case has more than two interacting entities, and at least two of them are nodes. The interaction represents the collective behavior (e.g., robots working together or interacting with humans). In this context, the entities can be nodes (e.g., robots) or humans.

We exclude projects that match *any* of the following exclusion criteria (EC):

Calls

EC1 Call started 2016–2017 so it contains no final results yet.

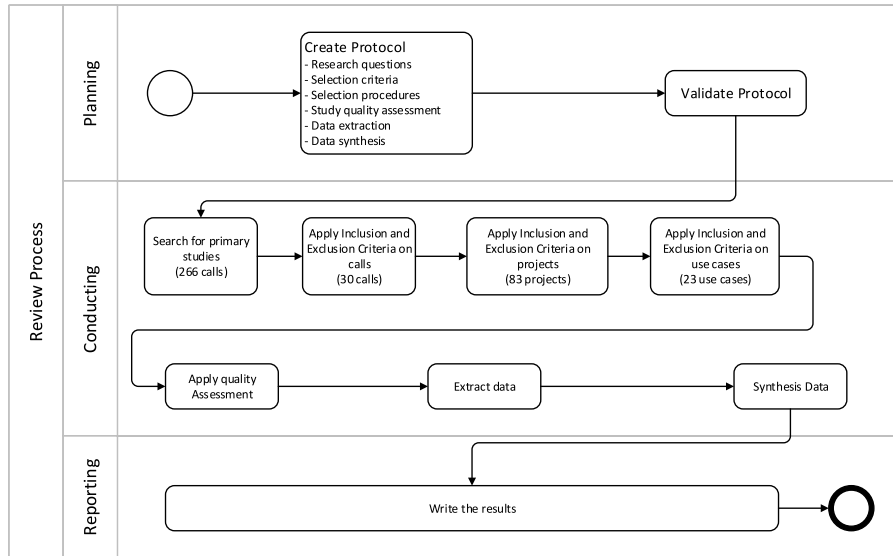


Fig. 1. The review steps.

Table 1
Selection criteria.

Criteria 1 Self-Adaptation asp.	Criteria 2 Cooperation asp.	Criteria 3 Physical asp.	Criteria 4 Entities
smart(*) intelligen(*) adapt(*) autonom(*) aware(*)	distribut(*) de(-)central(*) co(-)operat(*) communicat(*) collaborat(*) connective(*) emergent swarm collective	physical CPS Internet of Things IoT embedded device hardware robot	# > 2

(*) means zero or several letters, (-) means it can contain a hyphen in the word, (#) means number of entities (i.e., physical nodes, human, ...) that interact in the use case.

EC2 Call description contains only keywords in a context different from the one of the three first criteria (e.g., a *collaboration* between researchers).

Projects

EC3 Project description contains only keywords in a context different from one of the three first criteria (e.g., a *collaboration* between researchers).

EC4 Project does not have a working website—last time checked on 8/2/2018.

EC5 Project has no deliverables.

Use cases

EC6 Use case challenges do not target the Application Layer, nor the Service Middleware Layer mentioned in [14]. For instance, use cases that exclude target network reconfiguration at the link layer or collective knowledge exchange between people (e.g., social networks, software for rating places, roadmaps, and platforms).

EC7 Use cases have one-to-one or one-to-many relation between nodes such as robots and humans (e.g., wearables, tour guide robot).

4.4. Study selection procedures

We define the selection procedure following the guidelines in [4] for each call, project, and use case. To select the calls related to our focus in the study, we apply the inclusion criteria (IC1, IC2, IC3) and the exclusion criteria (EC1, EC2). We select a set of projects from the selected calls by applying inclusion criteria (IC4, IC5, IC6) and exclusion criteria (EC3, EC4, EC5). Afterward, we select the use cases in the selected projects by applying the inclusion criteria (IC7, IC8) and exclusion criteria (EC6, EC7).

4.5. Study quality assessment

We follow the selection procedures with an evaluation of the quality of the selected demonstrators. By answering the quality assessments, we get more accurate information about the demonstrators that serve the study focus. Valid answers for the following checklist items are: yes, partially, no. The list of related questions follows:

QA1 Is a challenge found in the deliverables missing from the project abstract?

QA2 Does the challenge in the deliverables require a collective behavior and involve uncertainties in the self-adaptation decision?

Table 2
Data extraction form for calls.

Data Items—Calls		
Call ID:	Call title:	Call description
Criteria 1:	Criteria 2:	Criteria 3:

Table 3
Data extraction form for projects.

Data Items—Project		
Call ID:	Project ID:	Project name:
Project title:	Description:	Website:
Criteria 1:	Criteria 2:	Criteria 3:

Table 4
Data extraction form for use cases.

Data Items—Use cases		
Project ID:	Use Case ID:	Use Case Description
Demonstrator:	Domain (RQ1):	Challenges (RQ2):
Criteria 4:		
Demonstrator Description: Deliverables and Reports:		
QA1:	QA2:	

Table 5
Data extraction form for uncertainty.

Data Items—Uncertainties (RQ3)		
Source(s)	Type(s)	Method(s)
CPS Part(s)	Self-Adaptation	Cooperation

4.6. Data extraction strategy

We use the extraction forms in [Tables 2, 3, and 4](#). Each table contains information needed to apply the selection procedures. The form for calls requires the title, the description of the call, and the fields for selection criteria 1, 2, and 3, which hold the data extracted from the project descriptions.

Similarly, the second form for project data requires the title, description, website availability, and the fields for selection criteria 1, 2, and 3.

The use-case-data form requires a description, domain, challenges, number of nodes, and the availability of documentation. At this stage, we target collecting the data that address RQ1 (i.e., “Use Case Domain”) and RQ2 (i.e., “Use Case Challenges”). In challenges, we need to focus on extracting the border situations that require adaptation.

Regarding the RQ3 (i.e., uncertainties), the related work presents an interest in sources of uncertainties, which in our case can be matched to CPS parts. However, the different types of uncertainties can arise from the same source, making it important to know the types of uncertainties in the system and identify the ones involved in the adaptation decisions and cooperation. The raw data with all the uncertainties usually do not take part in the adaptation decision without processing, which requires using methods that handle the identified uncertainties. Thus, we present a data extraction form to extract sources of uncertainties, types of uncertainties, and methods to handle uncertainties ([Table 5](#)).

4.7. Data synthesis

The data of the selected projects is extracted from the demonstrators. The synthesized data displays the existing relations between the domains and challenges in each demonstrator. Furthermore, it illustrates the related uncertainties in them. Moreover, many different classifications are synthesized from the extracted data demonstrating the information about the self-adaption.

Table 6
Selection of use cases.

Calls title [266]	Calls description [81]	Project title [384]	Project description [243]	Use cases [82]
S [81]	S [30]	S [243]	S [83]	S [23] O [59]
			O [160]	
	O [51]	O [141]		
O [185]				

S stands for Selected, O stands for Others.

“S[*]” is the selected number of items depending on the name of column.

“O[*]” is the not selected items depending on the name of column.

4.8. Approved protocol

The protocol was validated by three members of the Department of Distributed and Dependable Systems. The participants have experience in different areas of software systems, which include (1) Smart Cyber-Physical Systems and IoT, (2) Dynamic Program Analysis, Measurement Methodology, Performance Modeling, and (3) Analysis of Dynamic Languages, Interpolation-based Verification Techniques. For further validation from external reviewers in the community, the protocol was submitted and published as a position paper [17].

To enhance the quality of results, we excluded some keywords from the first keyword selection because the results proved too general for some of them. Also, we added the fourth criterion regarding the number of entities while requiring at least two of them as physical nodes because a demonstrator needs to show the collective behavior of at least two physical nodes making a decision. We also made slight updates to inclusion and exclusion criteria specifying the exact starting time of a project, the availability of documentation, and the layers targeted by the challenges.

5. Conducting the study

This phase concerns applying the steps decided in the planning to perform the review and get the results. In [Table 6](#), we illustrate the steps of the conducting with the resulted numbers of each selection procedure that starts with a total number of 266 of the presented calls in both FP7 and H2020.

After applying the selection procedures to calls using steps 1–8, the number of selected calls decreases to 30, corresponding to 384 projects. After applying steps 9–13, the number of the selected projects went down to 83 projects, corresponding to 82 use cases. Finally, after applying steps 14–15, the final number of selected use cases was 23.

Later, we applied the quality assessment on the resulted set of use cases. We ended up with the same number of projects and use cases. However, the number of challenges in the use cases was different. The indicators are interpreted as follows:

1. **QA1** evaluation determines which challenges to count in the selection. In other words, if the evaluation is “yes”, then the challenge is added. Otherwise, the challenge can be a part of another challenge, partially or totally. In such a case, the evaluation is “partially” or “no”, and the challenge is disregarded.
2. **QA2** determines which challenge contains uncertainties and requires collective behavior. If the evaluation is “no” or “partially” then disregard the challenge. The only evaluation to be considered is “yes”, which has both uncertainties and collective behavior. Then, the challenge counts.

We extracted the data from 12 selected projects and 23 use cases. To check the results of our protocol, we made an independent cross-check of the protocol on Horizon2020 calls of 2014–2015, and the extracted results were the same set of the selected use cases and challenges without the quality assessment.

6. Results

This section answers the research questions based on the data extracted from the selected demonstrators in EU projects. Each demonstrator is built upon a specific architecture, which plays a role in the adaptation and cooperation in the system. We first list the projects with a short description to explain their focus, then we step to answering the research questions and study the impact of uncertainties. To answer the research questions, we describe the situations that require self-adaptation in the selected demonstrators and their relation to system properties. Then, we explore the uncertainties in the demonstrators and study the methods used to handle these uncertainties. Based on the results in this section, we investigate the relations between the high-level and low-level aspects in the CPS design considering uncertainties. The results extract the links between the CPS Parts, Self-Adaptation, Autonomy Degree, and Uncertainties.

6.1. Projects

To explore the relations to uncertainties, we need to explore the presented architecture on which the demonstrators are based. Generally, the demonstrators in the projects use different system architectures that vary between interoperation, involvement of IoT, edge and cloud-based architecture, and distributed peer-to-peer architecture. To illustrate the variety of the architectures, we list high-level information about the projects in Table 7, which includes the selected demonstrators in our study that target collective behavior in sCPS (more details about each project can be found in [19]). The list of projects follows:

(T1) *VICINITY*. The project presents a standard independent user-centric platform that provides interoperability between different IoT platforms in different domains. It allows users to share information in these domains and the system to make better decisions in a decentralized way. This is achieved by using the concept of a virtual neighborhood network.

(T2) *sybiote*. The project aims at presenting a unified view of different IoT platforms. They can interoperate with each other in a secure way. The goal is to provide services for various smart environments, which contain sensors and actuators. Furthermore, these services are made available for end-users through mobile applications and web applications adaptive to the smart environment.

(T3) *CADDY*. The project aims at helping divers in working, orientating, and ensuring the diver's safety. The diver's physical state and behavior are monitored, and the information forms an input to the decision-making process, in addition to the environmental conditions.

(T4) *OrPHEuS*. The project presents collaborative hybrid-grid control strategies to minimize the waste in photovoltaic (PV) electricity production. The proposed solution considers new control strategies to transform overproduced electricity to energy for heating water or surfaces.

(T5) *Clout*. The project introduces dynamic storage of historical data in a cloud that allows users to share services. Since the size of stored data from city IoT is unpredictable and huge, elastic and scalable clusters are required. On the other hand, sharing resources with many users at the same time may cause conflicts, which is also addressed by this project.

(T6) *RECONFIG*. The project focuses on formal method-based planning for collaborative grasping and carrying objects by robots. The objectives of the projects are to improve dealing with visual information, robot-to-robot gesturing detection, point-to-point navigation, localization, stable grasp, and collaborative picking and carrying of items.

(T7) *hydrobionets*. The project aims at developing Wireless BioMEM Networks (WBNs), which have microbiological awareness and are used for distributed monitoring of the water treatment and desalination process in large-scale industry settings to improve productivity.

(T8) *INERTIA*. The project presents a system to manage the efficiency of energy smart grids. It takes into account user comfort and provides dynamic pricing that minimizes expenses.

(T9) *NIFTi*. The project targets cooperation between humans and robots to rescue people. Urban Search and Rescue (USAR) involves Human-Robot Interaction (HRI) by spoken dialog and graphical user interface.

(T10) *Greencom*. The context of this project is Smart Grids and Energy. It aims to assure the users' comfort on the one hand and save energy consumption on the other one. This is done by monitoring and supporting the decision-making of customers. Hence, the targeted challenge lies in having efficient energy management that enhances performance while preserving security.

(T11) *ASCENS*. The project targets designing dynamic systems with collective behavior. Those systems aim to form a group able to solve complex problems that an individual cannot. Therefore, a methodology that targets emergent behavior engineering is presented. It focuses on self-adaptation and awareness issues in addition to performance.

(T12) *IPAC*. The project presents a platform for building collaborative autonomic embedded systems on mobile nodes. The proposed middleware is used for developing services that are context-aware and use short-range communication for information exchange (i.e., event-based). Thus, the used information dissemination algorithms are similar to how rumor spreading works.

6.2. Domain classification

RQ1: What are the typical domains for collective sCPS?

The variety of domains in the selected demonstrators ensures covering multiple options of the used architecture and addressed goals. Thus, we illustrate the domains that the selected demonstrators fall into following the classification presented in Table 8. The classification extends another classification that is presented in [20] (i.e., Smart Manufacturing, Emergency Response, Intelligent Transport, Robotics for Service, ...). Although other publications presented a classification of domains, this publication targets studying the challenges for CPS applications similarly to the first two research questions in our study, which are essential to answer. Thus, we decided to base our classification partially on this publication and apply the required modifications when necessary. It is worth mentioning that there is an overlap between these domains; in our case, however, we relate each use case to just one domain. We list the domains of the selected demonstrators and their description as follows:

(D1) *smart factory and manufacturing*. This domain includes all the smart industry aspects, from smart technologies in production to manufacturing products. Furthermore, smart industries and warehouses use monitoring, tracking products, and quality control of production processes (e.g., monitoring water/gas/oil distribution network). Therefore, industry automation increases efficiency and reduces costs. In addition, this includes manufacturing new products and their customization, such as new boards and energy batteries.

Table 7

Projects of selected use cases.

Project name	Project title
T1: VICINITY (ID: 688467) (1.1.2016–31.12.2019)	Open virtual neighborhood network to connect intelligent buildings and smart objects. Website: http://vicinity2020.eu/vicinity/
T2: symbloTe (ID: 688156) (1.1.2016–31.12.2018)	Symbiosis of smart objects across IoT environments. Website: https://www.symbiote-h2020.eu/
T3: CADDY (ID: 611373) (1.1.2014–31.12.2016)	Cognitive autonomous diving buddy. Website: http://caddy-fp7.eu/
T4: OrPHEuS (ID: 608930) (1.12.2015–31.5.2018)	Optimising Hybrid Energy grids for smart cities. Website: http://www.orpheus-project.eu/
T5: ClouT (ID: 608641) (1.4.2013–30.5.2016)	ClouT: Cloud of Things for empowering the citizen cloud in smart cities. Website: http://clout-project.eu/
T6: RECONFIG (ID: 600825) (1.3.2013–31.5.2016)	Cognitive, Decentralized Coordination of Heterogeneous Multi-Robot Systems via Reconfigurable Task Planning. Website: http://www.reconfig.eu/
T7: HYDROBIONETS ^a (ID: 287613) (1.10.2011–31.12.2014)	Autonomous Control of Large-scale Water Treatment Plants based on Self-Organized Wireless BioMEM Sensor and Actuator Networks. Website: http://www.hydrobionets.eu/
T8: INERTIA ^a (ID: 318216) (1.10.2012–30.9.2015)	Integrating Active, Flexible and Responsive Tertiary Prosumers into a Smart Distribution Grid. Website: http://www.inertia-project.eu/inertia/
T9: NIFTi (ID: 247870) (1.1.2010–31.12.2013)	Natural human–robot cooperation in dynamic environments. Website: http://www.nifti.eu/
T10: GreenCom ^a (ID: 318213) (1.10.2012–30.4.2016)	MyGrid; Energy Efficient and Interoperable Smart Energy Systems for Local Communities. Website: http://www.greencom-project.eu/
T11: ASCENS (ID: 257414) (1.10.2010–31.3.2015)	Autonomic Service-Component Ensembles. Website: http://www.ascens-ist.eu/
T12: IPAC ^a (ID: 224395) (1.5.2008–28.2.2011)	Integrated Platform for Autonomic Computing. Website: http://ipac.di.uoa.gr/

^aIn case the project website is not accessible after the date in the protocol, it is possible to access them using websites such as <https://timetravel.mementoweb.org/> or <http://web.archive.org/>.

Table 8

Domain examples.

Domain	Example
D1: Smart Factory and Manufacturing	Automation of monitoring and tracking products, manufacturing boards
D2: Smart Traffic and Transport	Smart parking, navigation
D3: Smart Homes and Buildings	Efficient energy and gas use
D4: Smart Grids and Energy	Decrease energy waste, renewable resources
D5: Robotics	Cognitive robots, monitoring robots
D6: Crisis and Emergency	Rescue unmanned robots
D7: Cloud	Service and resources provider

(D2) *smart traffic and transport*. This domain consists of two basic parts: (semi-) autonomous vehicles/plains/trains and smart management of traffic. Automation in vehicles allows them to navigate, self-drive and smart-park, while traffic management provides an infrastructure for predicting traffic and support for smart highways.

(D3) *smart homes and buildings*. This domain includes the automation of urban spaces such as homes, offices, and buildings. Automation requires having a communication infrastructure known as Internet of Things (IoT). It consists of a network of sensors and actuators for monitoring and decision-making purposes. The aim is to ensure optimized and efficient use of resources (e.g., energy) in addition to safety.

(D4) *smart grids and energy*. This domain describes the management of smart grids to keep the balance between the demand and the generated energy. Furthermore, renewable energy is the future technology where the electricity is generated from environment-friendly sources, and the system raises the customer's awareness about his energy consumption and provides him with the ability to customize the services.

(D5) *robotics*. This domain contains developing robots that are cognitive or mimic the behavior of living organisms and robots that cooperate with humans. For instance, a robot monitors the environment to help humans in navigation underwater.

(D6) *crisis and emergency*. This domain contains the devices and robots that help in case of crisis by forming teams with humans in rescuing tasks.

(D7) *clouds*. This domain presents cloud providers for services and resources (i.e., software and hardware) for IoT entities and users.

The results in Fig. 2 show the number of selected use cases in the projects in each domain. The highest number of the use cases are under Smart Traffic and Transport, and Smart Grids and Energy.

6.3. Challenge classification

RQ2: What are the challenges addressed by collective sCPS?

The challenges that selected demonstrators address can be classified under non-functional requirements as it is presented in [20] (system properties). We used the same concept and grouped the challenges under a set of properties, which we present in Table 9. In our case, we care about the challenges that reflect border situations in the system, which require the system to adapt to satisfy the requirements. We define the properties associated to the challenge groups that are tackled by the demonstrators in this study as follows:

(P1) *safety*. This property includes the challenges that target avoiding dangerous situations for humans or unwanted circumstances in a collective manner, such as dangerous health conditions [T3] and dangers from being in unexpected environments such as detecting intruders [T2]. Another example is avoiding vehicle accidents [T12].

(P2) *performance*. This property targets the challenges related to managing resources in the system. More specifically, the resources include many aspects such as hardware, time, and energy. For instance, energy management [T1][T2][T4][T8][T10][T11] is an important issue, especially in Smart Grids since providing a better demand and response strategies can hugely help in decreasing energy waste [T4]. Additionally, load balancing and sharing resources in the case of the cloud can improve performance and scalability [T11]. Also, issues related to utilizing the buildings and parking slots usage [T1] are considered, in addition to optimizing routes. Tracking the quality of a service or a product is tackled in [T7] that considers the water treatment during the desalination process and [T12] during the production process.

(P3) *price*. This property targets the challenges that consider the financial aspect of the demonstrator. Indeed, applying any new technology in real life has the hardware cost, the installation, and the running as one of the main concerns; therefore, it should be reasonably balanced with the other properties. For instance, using renewable energy generation minimizes the cost of energy generation [T1], smart energy management [T10], and transforming the energy to different forms to reduce waste [T4]. Also, utilizing resources reduces the costs, such as stopping Virtual Machines (VM) after being idle for a while [T11].

(P4) *resilience*. This property includes the challenges that target dealing with failures of some components [T2][T3][T11], abnormal behavior [T7], resolving conflicts [T5], and handling unexpected or manipulative situations by increasing system awareness to its context [T6]. Thus, it requires that the system features a level of dynamicity that allows it to adapt to the current circumstances that might include humans as a part of the system [T10]. Most importantly, this dynamicity mainly requires a collective behavior to overcome or adapt to the context changes and recover from the failures so that the system still provides the services to the end-users.

(P5) *cooperation*. This property includes the challenges that target the group behavior of system components, which aims at providing required services. For instance, robots exchange information to achieve shared goals such as moving objects [T6], rescue missions [T9] (with human involvement [T11]), avoiding collisions [T9], finding parking lots [T12], and robot–human collaboration in diving [T3].

(P6) *connectivity*. This property includes the challenges that target communication problems, including limitations in communications [T6], short ranges [T7], and network problems [T12]. Hence, the system tries to handle those issues collectively.

(P7) *authorization*. This property includes the challenges that target authority issues in the systems to avoid criminal actions such as hacking inside a home sensor network that could cause privacy violations and hacking related to vital resources management systems [T10]. Additionally, handling physical access issues to facilities such as ports is also considered [T2].

The results in Fig. 3 show that the most targeted properties in collective sCPS are Performance and Resilience.

6.4. Classifications of uncertainties

RQ3: How uncertainties relate to runtime adaptation decision in collective sCPS?

To answer this question, we need to understand from where the uncertainties emerge in CPS parts (i.e., sources of uncertainties), the types of uncertainties that exist in the collected system data, and which methods to use to handle the uncertainties before performing adaptation decisions. The extracted information from the study will provide us with links between CPS parts, adaptation, and uncertainties.

To present the results, we classify the sources of uncertainties based on [21], which is a study investigating the sources of uncertainties in self-adaptive systems thoroughly on different levels of the design process. The publication is highly related to our study since it also presents the theories used to handle uncertainties, uncertainties types associated with the MAPE-K loop, and how to model uncertainties. However, the presented methods are limited to statistical and probabilistic approaches, and the reasoning presents the relations between the sources of uncertainties, types of uncertainties, and how to model them. However, the classification is meant to be used and extended as needed.

Sources of uncertainties classification. We start the investigation with presenting the sources of uncertainties in the selected demonstrators as following:

Adaptation functions uncertainty This class of sources of uncertainties is related to uncertainties caused by variability, scenario, and runtime uncertainties.

(S1) *variability space of adaptation*. This source is about the size of variability space involved in the adaptation decision, where it is problematic to select the most relevant variable from a huge variables space, such as monitoring human physical state (i.e., vital parameters) [T3].

(S2) *effecting*. This source of uncertainty is related to unpredictable behavior of effectors that can be caused by a system error, which can lead to abnormal behavior and malfunctions [T7].

(S3) *automatic learning*. Here, the source of uncertainties is related to errors from the used technique in making decisions or applying inadequate models used to represent the data, such as machine learning and statistical approaches, which include

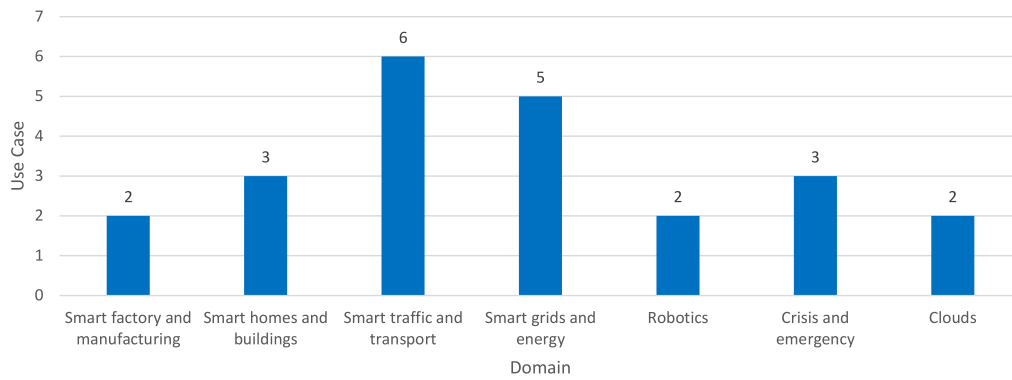


Fig. 2. The numbers of use cases in each domains.

Table 9

Properties and challenges.

Property	Challenge	Border situation	Example
P1: Safety	Dangers from unexpected environment	Human body functions failure	Detect human physical state [T3]
P2: Performance	Optimize resources usage and learn context and human behavior	Lack of resources or inefficient resource management	Hybrid energy grid [T4], Sharing resources in the cloud [T11]
P3: Price	Smart management for resources	Inefficient resources management	Add renewable energy generators [T1], Minimize cost by learning from tariff and resources consumption [T10].
P4: Resilience	Overcome failures	Component Failure	Sensors classification faults statistically [T5], Overcome failure of an individual by forming new ensemble [T11], Spotting new chair in the room that was not there [T6]
P5: Cooperative	Unexpected and changing context and goal achievement requirements	Manipulated situations and inefficiency in achieving the goal by individual	Robot–human collaboration in diving [T3], Rescue mission by a team of robots [T11]
P6: Connectivity	Limited Communication with collisions	Communication limitations	Short range of communication between sensors in the industry [T7]
P7: Authorization	Authorization for accessing physical facilities or data	Unauthorized access or manual process	Authority of multi-sensory network in homes [T10]

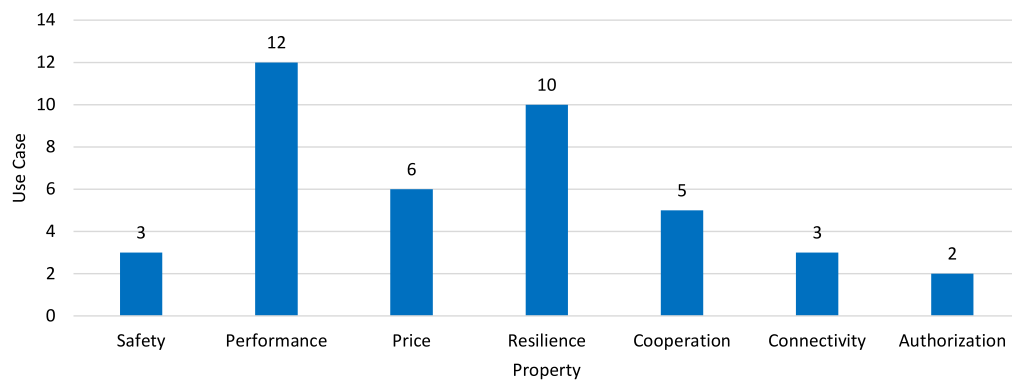


Fig. 3. The numbers of use cases for each property.

some amount of errors during their calculations to find the fitting models. For instance, machine learning and statistical approaches are used for efficient resources and cost management [T1][T2][T4][T8][T10][T11].

(S4) *decentralized*. This source of uncertainties is related to entities taking decisions collectively, which is sensitive to the frequency of data exchanging based on the scenario to keep the data updated for adaptation decisions, such as exchange information between vehicles to avoid accidents [T12].

(S5) *changes in adaptation mechanisms*. The source of uncertainties is related to changing the infrastructure according to changing the goals, which impacts the qualitative attributes in the system, such as regulation of room temperature is subject to the heating strategy [T2].

(S6) *fault localization and identification*. The source of uncertainties is related to noisy measurements or identifying faults. We added the noisy measurement to this part since the parts are mostly related to localization or identifying objects, such as robot position measurements [T3].

(S7) *human in the loop*. This source of uncertainties is related to human behavior when he/she is involved in the decision-making process, such as Robot–Human cooperation to perform a rescue mission [T9], which means the human can make errors during the mission as well.

Environment uncertainty This class of sources of uncertainties is related to uncertainties caused by execution context and human behavior.

(S8) *execution context*. This source of uncertainties is related to the unexpected environment, which provides a context that the components are not aware of causing possible issues, such as robots collisions [T9].

(S9) *multiple ownership*. This source of uncertainties is related to conflicts caused by having multiple sources of information, such as conflicting tasks for a robot [T3].

Resources uncertainty This class of sources of uncertainties is related to uncertainties that arise from system resources.

(S10) *changing resources*. The source of uncertainties here is related to the dynamicity of resources, such as resources loss, as in the case of the energy grid [T4].

The results in Fig. 4 show that the most common source of uncertainties in the selected demonstrators are Automatic Learning and Changing Resources.

Types of uncertainties classification. In addition to the sources of uncertainties, we need to recognize the types of uncertainties that the use cases include. Knowing the types of uncertainties help in identifying how to model them during design time before being handled by the methods, which can have assumptions that correspond to the chosen models of uncertainties. Even though we do not tackle modeling uncertainties in this study, we consider the assumptions of the methods that handle the uncertainties, as we will discuss later. We divided types of uncertainties into categories as follows:

(U1) *network and delays*. This category is related to end-to-end network communication problems [T4][T6][T9][T12], such as delays [T6][T12] and offloading latency [T4], in addition to cases of having limited [T9] or no direct communication [T9] (i.e., we consider the noise in the communication as a separate type).

(U2) *missing information*. This category is related to lack of knowledge [T4], losing information [T9][T12], lack of detection [T2], and missing information due to malfunction of a part of the system [T3][T7].

(U3) *noise*. This category is related to noise in measurements [T2], communication [T7] or processing noisy images [T3], inaccurate localization [T3][T6][T9][T12] and pointing or observation [T3][T6].

(U4) *violating operational boundaries*. This category is related to keeping the values under a certain threshold. For instance, efficient resource management such as energy [T1][T2][T8], road planning [T2], parking space [T1], and hardware resources [T11] require dealing with peaks in the usage, also keeping the efficient cost management of used resources [T1][T4][T10][T11].

(U5) *ambiguity and ill-definition*. This category is related to problems in defining the models such as bacteria functioning [T7], recognizing bad gestures [T3] or objects [T6], the risk from dynamic features as in heart attacks [T3], and sensors fault classification [T5].

(U6) *failures*. This category is related to failures of a component/part of the system or abnormal behavior [T2][T3][T5][T6][T7][T11].

(U7) *inconsistency*. This category is related to verifying the local evaluation [T12] and avoiding collisions [T3][T7][T9] by resolving inconsistency or conflicts.

(U8) *other uncertainties related to context*. This category is related to photovoltaic (PV) energy surplus/overvoltage [T4], unknown environment [T9], flexibility in energy management by context awareness [T4][T10], such as in case of detecting the comfort level for occupants in a room [T1][T8], and mental occupancy of a rescuer [T9], and avoiding obstacles [T11].

The results in Fig. 5 show that most of the selected demonstrators have Violating of Operational Boundaries, then Network and Delays and Failures.

Methods to handle uncertainties classification. We care about the methods because they are in between handling uncertainties in the input data and the outputs that might be used in another method or take part in the adaptation decisions directly. To explore that, we first classify the used methods to handle uncertainties in the selected demonstrators in the projects (Table 10) as follows:

(HU1) *data filtering and estimation*. This group contains three basic categories, which are a projection of the variable space to a smaller dimension (U3 - T7), estimation such as in the case of data fusion (U3, U7, U8 - T9) (U2, U3, U7 - T12) interpolation (U4 - T2)(U2, U6 - T2) or filtration (U1 - T6)(U1 - T9), and filling data such as matrix completion (U2 - T7).

(HU2) *statistics and probability*. This group contains frequencies (U4, U8 - T1)(U8 - T4)(U4, U8 - T8)(U3, U7, U8 - T9)(U4, U8 - T10), such as (S)ARMA (U5, U6 - T5), probability, such as fuzzy logic (U5, U7 - T3), and Markov processes (U5, U6 - T5)(U3, U6 - T7)(U4, U8 - T8).

(HU3) *(re-)configuration*. This group is divided into two parts: configuration at design time, such as adding more sensors to increase accuracy (U3 - T3), and reconfiguration at runtime (U3, U6 - T2)(U3, U5 - T3), such as dynamic communication interfaces (U4 - T4)(U1 - T12) or deactivating a malfunctioning sensor (U6 - T7).

(HU4) *machine learning and neural networks*. This group contains unsupervised and supervised learning such as neural network (U1, U3, U5, U6 - T7)(U3, U5 - T6) and SVM (U4 - T1)(U4 - T2)(U3, U5 - T3)(U3, U8 - T9)(U4, U8 - T10), and reinforcement learning (U1, U2, U4, U8 - T4).

(HU5) *verification*. This group contains design time and runtime verification, such as SMC-BIP statistical model checking (U6 - T11).

(HU6) *human-in-the-loop*. This group contains total involvement of humans in the decisions, such as notify users about their energy consumption (U2 - T2)(U4 - T10), human giving orders by gestures (U3, U5 - T3), and partial involvement such as robot–human cooperation to perform rescue mission (U2 - T9).

(HU7) *declarative programming*. This group contains constraint programming, such as ensembles of entities that are formed based on constraints (U4, U6 - T11), modeling such as behavioral trees that mathematically models the execution of tasks (U3, U6 - T6)(U6 - T11), and logic programming such as Prolog that uses facts and rules to infer information (U1 - T12).

The results in Fig. 6 show that the majority of the selected demonstrators include Statistics and Probability, Machine Learning, and Neural Networks as methods for handling uncertainties.

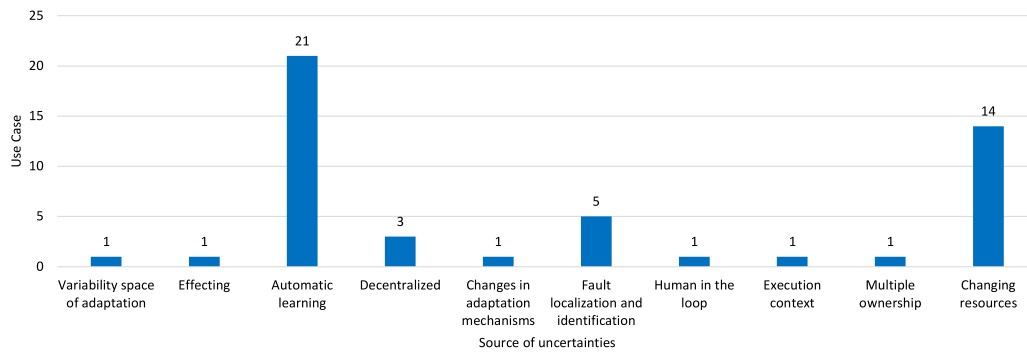


Fig. 4. The numbers of use cases for each source of uncertainties.

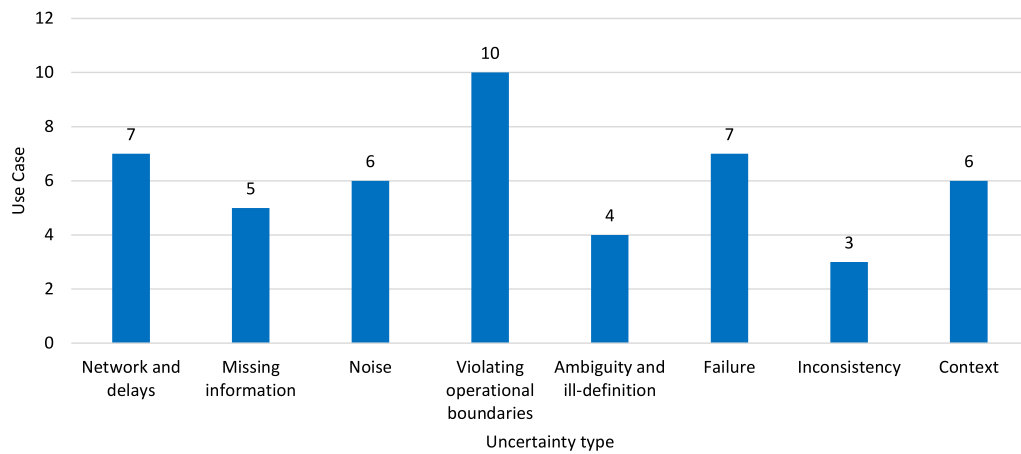


Fig. 5. The numbers of use cases for each type of uncertainties.

Table 10
Methods classification.

Method classification	Technique example	Method example
HU1: Data Filtering and Estimation	Projection Estimation Filling Data	Subspace Projection [T7] Kalman Filter [T6] Matrix Completion [T7]
HU2: Statistics and Probability	Frequencies Probability Markov process	(S)ARMA [T5] Fuzzy Logic [T3] Hidden Markov Model, Semi-Markov [T7]
HU3: (Re-)Configuration	Design Time Runtime	Static adding for more sensors during the design time [T3] Dynamically change of communication interfaces [T12]
HU4: Machine Learning & Neural Networks	Supervised Learning	Neural network [T7]
HU5: Verification	Runtime	SMC-BIP[T11]
HU6: Human-in-the-loop	Total Involvement Partial Involvement	Human takes over: User Notification/Alarm [T10] Cooperation with human [T9]
HU7: Declarative Programming	Constraint programming Modeling Logic programming	Ensemble (i.e. CSPs) [T11] Behavioral Trees [T6] Prolog [T12]

Links between uncertainties and collective sCPS. The sources of uncertainties exist in different CPS parts; the relations we extracted from the CPS parts that cause the uncertainties are ordered as follows: Cyber part, Physical part, Network part, Environment part. To extract more detailed information related to self-adaptation, we approached the self-adaptation aspect, which presents an adaptation taxonomy based on a list of questions proposed in [22]. The questions that any adaptive system should be able to answer are: *When* to adapt? *Why* do we have to adapt? *Where* do we have to implement a change? *What* kind of change is needed? *Who* has to perform the adaptation? *How* is the

adaptation performed? Answering the previous questions provide many details about the system architecture, such as autonomy degree that reflects the collective behavior (*How?*), computation level (*Who?*), and adaptation techniques (*What?*) (more details can be found in Appendix Table A.14).

From the self-adaptation questions and based on the study focus, we decide to choose the *adaptation techniques* (i.e., What—parameter, structure, context) for its relation to performing adaptation decisions and the *Autonomy degree* (i.e., How—decentralized, semi-centralized and centralized) for its relation to cooperation between the system components to satisfy system properties.

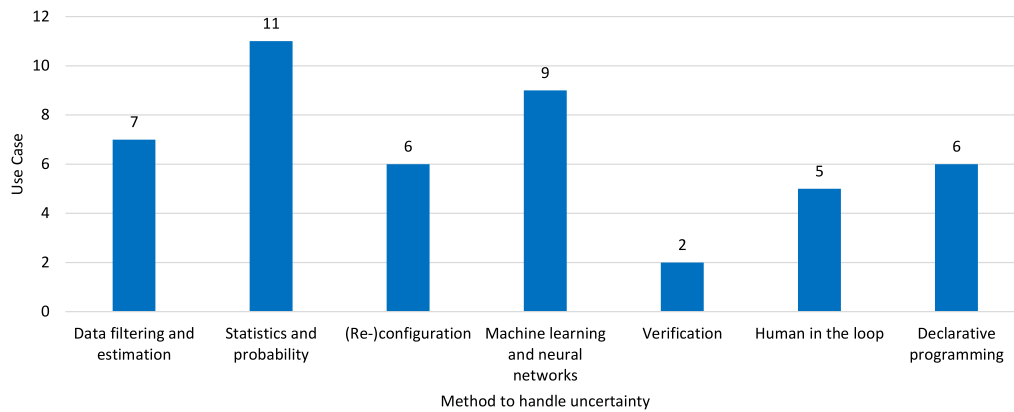


Fig. 6. The numbers of use cases for each method to handle uncertainties.

Both choices are associated with system architecture, and it is interesting to see the relation to the uncertainties and the methods to handle them.

We counted the number of methods based on CPS Part, Adaptation Technique, and Autonomy Degree. It is worth mentioning that the CPS Parts and Adaptation Techniques are not mutually exclusive. The results in Fig. 7 show that the selected demonstrators have roughly the same number of the types of uncertainties and the methods to handle, which indicates that, on average, each kind of uncertainty is handled by a method. However, some cases require more than one method to handle a specific type of uncertainties, or one method to handle multiple types of uncertainties. From the CPS part perspective, the Cyber parts, as a source of uncertainties, is the most targeted category. From the adaptation techniques perspective, the parameter technique is the most targeted category. Finally, from the autonomy degree, the demonstrators widely support the decentralized manner, and in some cases, they switch to semi-centralized or centralized decision-making.

To this end, the extracted data from the study presents different dimensions of the selected demonstrators. However, we need to reason about the relations between the high-level and low-level aspects in the design using the observations over the extracted data.

7. Reasoning over the results

After answering the research questions, we aim at synthesizing the relations between the uncertainties and several collective sCPS aspects.

We first relate the demonstrators in different domains with the challenges and the types of uncertainties. From the demonstrators and the mapping, we observe many possible relations between the uncertainties and the architecture. From these observations, we introduce a taxonomy for the methods to handle uncertainties. The inferred relations from the observations are not certain; we represent the results using a Bayesian Network-based framework on Multi-Criteria Decision Making [23]. The representation allows the developers to experiment with the results and investigate multiple options for their design.

Based on the previous synthesis, we propose a framework for the developers to help them during the design stage by focusing on the uncertainties and their impact on adaptation decisions. The guide starts from the requirements and ends up with uncertainty-aware adaptation decisions using some insights from the synthesis. Finally, we demonstrate the usage of the results on an example from the projects.

7.1. Observations over mapping the extracted data

From the classifications and simple listing of the data, we come to important conclusions regarding the most targeted categories in each aspect. However, the complexity of the CPS requires more elaborated investigation to interrelate different aspects of CPS design.

Thus, we start by illustrating (Table 11) the relation between each property and use case. If there are more than one use case in the same domain in the same project, we merge them in the table.

The table contains the targeted challenges with the existing uncertainties mentioned by the demonstrators (# is used if there is a challenge but no collective behavior or/and uncertainties mentioned). The results show that in Robotics and Crisis & Emergency domains, the research does not focus on performance, cost, or authorization issues, while the rest of the domains take performance and cost into account as two of the main challenges. The performance and cost are usually coupled together, making the challenge mostly related to realizing the context and managing resources. In Robotics and Crisis & Emergency, the focus mainly lies on resilience and cooperation to achieve a common goal, where failures are the main issue. Connectivity has network problems and delays as the main challenge, while safety challenges vary between delays, inconsistency, and ambiguity. Later on, we skip mentioning the authorization since there was no collective behavior in addressing the challenge.

We notice many aspects from Table 11 and the answers of the research questions that are candidates to form a relation between uncertainties and architecture, which are:

Domains. From Table 11, we conclude that the domain impacts the property of interest and the types of uncertainties and methods. For instance, in Robotics and Crisis & Emergency domains, we observe that the demonstrators are directed toward achieving goals rather than improving performance. We can notice that the types of uncertainties are handled thoroughly with multiple methods to improve the data quality for better decisions at runtime. However, the best practices by the developers in the domain can also play a role. For instance, in the Smart Factory & Manufacturing domain, uncertainties tend to be handled with the least effort possible, e.g., by deactivating a malfunctioning sensor and monitoring entities that have neither models nor assumptions (T7-HYDROBIONETS).

Properties. Looking into the relations between properties in Table 11, we notice that a property can be coupled with another property in a trade-off relationship. However, the relations of these trade-offs to types of uncertainties and methods is

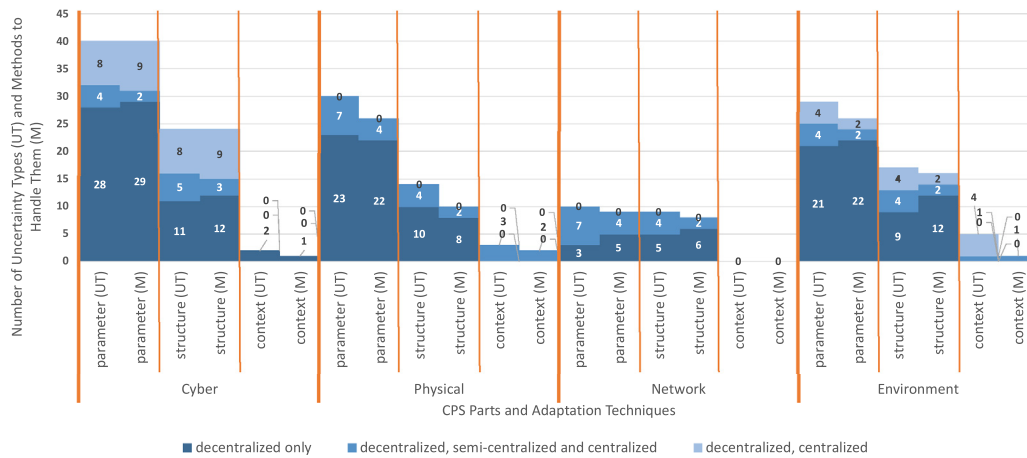


Fig. 7. The numbers of Uncertainty Types in use cases for each CPS Part, Adaptation Technique, and Autonomy Degree.

not clear and worth investigating. For instance, having performance in mind, the noise is handled in resilience using sub-space projection or in connectivity using more measurements (T7-HYDROBIONETS).

Design/runtime. Most of the uncertainties are handled by methods at runtime (Table 10). Nevertheless, there are design time solutions related to noise, ambiguity, or delays. Simply, the solutions are in deciding upon the kind and number of sensors involved in monitoring or the type of communication (e.g., broadcasting) (T12-IPAC). These solutions are extended in other use cases to runtime reconfiguration solution (e.g., communication interfaces reconfiguration).

Methods and assumptions. The same methods that are used in general cases could be enhanced by adding different parameters (i.e., methods classification). For instance, it is better to use a trajectory instead of a position and speed when following a robot, i.e., not relying on communication but visual recognition or choosing a special combination of sensors to enhance the accuracy of measurements. Another case is overcoming model limitations by combining the model with another one. For instance, it is possible to predict occupancy depending on historical data by creating a Transition Matrix based on Markov Chains (MC) (since MC has no memory) (T8-INERTIA). Finally, the selection of the method can depend on the required inputs as training data for machine learning or on expected outputs from the method such as classification.

Level to perform computation. The system can provide resources for the computation of the methods based on the available infrastructure, which can involve a different set of uncertainties and require re-selection of the methods used to handle them (i.e., methods classification). This selection impacts designing the system, such as handling noise and ambiguity, which could be made on the local node during recognizing trajectory (e.g., T6-RECONFIG). Choosing a communication over Distributed/Peer-to-Peer (D/P2P) architecture implies that delays should be handled in addition to other possible uncertainties (e.g., T9-NiFTi).

Human intervention and degree of autonomy. Some domains prefer to keep human involvement, such as in Crisis & Emergency for psychological reasons (i.e., avoid designing fully autonomic robots because of trust issues—Table 11; T9-NiFTi).

Relations between uncertainties. For the developer, it is not enough to be aware of the types of uncertainties that can influence the chosen system property (i.e., methods classification

and Table 11). They also need to consider the relations between them. For instance, safety property is influenced by delays in D/P2P architectures (i.e. T12-IPAC). This uncertainty is related to inconsistency in data, which also influences safety. On the other hand, it is related to missing information, and both influence performance. In case of delays, the approach to handle it is to reconfigure the communication interface, which also helps handle inconsistencies. For performance, the solution is based on using broadcasts to minimize the delays and handle missing data by lowering the channel sampling. Another example is related to the performance property. The first case is to handle delays and missing information alone in D/P2P level (i.e., T12-IPAC), while other cases are delays, missing information, threshold, and the context in edge level (i.e., T4-OrPHEuS). In the first case, the delay is handled by architectural decisions using broadcasting, while in the second case by using Stream Processing Framework. While the missing information in the first case is minimized using lower channel sampling, in the second case, interpolation and prediction are used because threshold and context are also involved in the use case. In other words, dealing with peaks needs a model of values for the evaluation. Choosing the other solution (i.e., lower channel sampling) means fewer values, impacting the model and the final evaluation.

Looking into the observations, we can conclude that the uncertainties can be influenced on too many levels. However, a further investigation about the strength of influence is required.

7.2. Representing the results in Bayesian networks and multi-criteria decision making

From the previous conclusions over the systematic review, we learned that many factors influence the uncertainties and the selection of the methods to handle them. Thus, we decided to represent a set of factors as a diagram that reflects the relations between them. We aim at presenting a semi-automatic way to select the methods to handle uncertainties supported by the collected data.

Since the sample of the projects is relatively small, and the influences are uncertain, we decided to use the Bayesian network-based Multi-Criteria Decision Making [23]. The study presents a combination of Bayesian networks and influence diagrams to represent the possible selection of multiple criteria and capture the uncertainties in the relations between the nodes. *Bayesian Network* is defined as an acyclic directed graph (DAG) that captures the qualitative part through nodes, while the quantitative part is presented through joint probabilities. This structure provides a

Table 11
Selection of use cases.

Domain	Project name	Safety	Performance	Price	Resilience	Cooperation	Connectivity	Authorization
Smart Factory and Manufact.	HYDROBIONETS		U5 HU2		U2,U3,U6 HU1,HU3		U1,U3,U6 HU2,HU4	
	IPAC		U1,U2 HU1,HU3			U3 HU3		
Smart Homes and Buildings	VICINITY		U4 HU2,HU4	U4 HU2				
	sybloTe	#	U4 HU1		U2, U3, U6 HU3,HU6			
Smart Traffic and Transport	VICINITY		U4 HU2,HU4, HU6	U4 HU2				
	sybloTe		U4 HU4		U2,U6 HU1			#
	ASCENS		U4 HU7					
	IPAC	U1,U7 HU3, HU7						
Smart Grids and Energy	OrPHEuS		U1,U2,U4,U8 HU2,HU4	U4 HU3				
	GreenCom		U8 HU2, HU4	U4 HU2, HU4, HU6	U8 HU2,HU4			#
	INERTIA		U4,U8 HU2					
	VICINITY		U4,U8 HU2	U4,U8 HU2				
Robotics	CADDY	U5 HU7			U3,U5,U7 HU4,HU6, HU7	U3,U5,U7 HU1,HU3, HU4,HU5,HU6		
	RECONFIG				U3,U6 HU2,HU4, HU7	U3,U5 HU2,HU4	U1 HU1	
Crisis and Emergency	NiFTi				U3,U7,U8 HU1,HU2	U1,U2,U3,U8 HU1,HU2, HU4,HU6		
	ASCENS				U6,U8 HU5,HU7	U6 HU5,HU7		
	IPAC						U1 HU3	
Clouds	ClouT				U5,U6 HU2			
	ASCENS		U4 HU2,HU7	U4 HU2, HU7	U4,U6 HU7			

Normal font: before QA1, Bold font: after QA2, #: no uncertainties.

suitable way to represent the aspects we need as nodes and put the collected data into use as probabilities. The influence diagram presents a method to make decisions, which requires adding a decision and utility nodes. The decision nodes hold the alternative solutions for the problem at hand, while the utility nodes hold the utility of the outcomes (see Table 12).

For our case, we used the GeNIe tool to create the Bayesian Network, where we define the network to contain the following parts:

The results in Fig. 8 show that the use case domains and system properties have a firm relation to the types of uncertainties comparing to their relation to the methods to handle the uncertainties. The reason is that the types of uncertainties are related

to the use case specification, unlike the methods proposed to handle selected uncertainty types, which are not domain-specific as strongly as the types of uncertainties. The autonomy degree and adaptation techniques have a strong relation to the properties and methods, suggesting that the choice of methods should be aligned with the autonomy degree and adaptation techniques used in the system. The CPS parts are also related to properties and sources of uncertainties. We can see that the relation between the output and the method is stronger than the inputs and assumptions regarding the methods. It is worth mentioning that the current representation assumes independence between the values of each node. If we aim to study their dependency, each

Table 12
Classification of variables in collective sCPS design.

Uncertainty awareness during collective sCPS design	
Objectives	To get information about uncertainties in similar demonstrators considering aspects of CPS Parts, Adaptation Technique, and Autonomy Degree
Decision problem	To determine the most suitable method to handle a specific uncertainty
Set of possible actions	The alternatives are the domains, where the choices of the domain are (Smart Factory and Manufacturing, Smart Homes and Buildings, Smart Traffic and Transport, Smart Grids and Energy, Robotics, Crisis and Emergency, Clouds)
Criteria	Property, CPS Part, Adaptation Technique, Autonomy Degree
Constraints	–
External factors (uncontrollable variables)	Uncertainty Source, Uncertainty Type
Internal factors (controllable variables)	Input, Output, Assumption

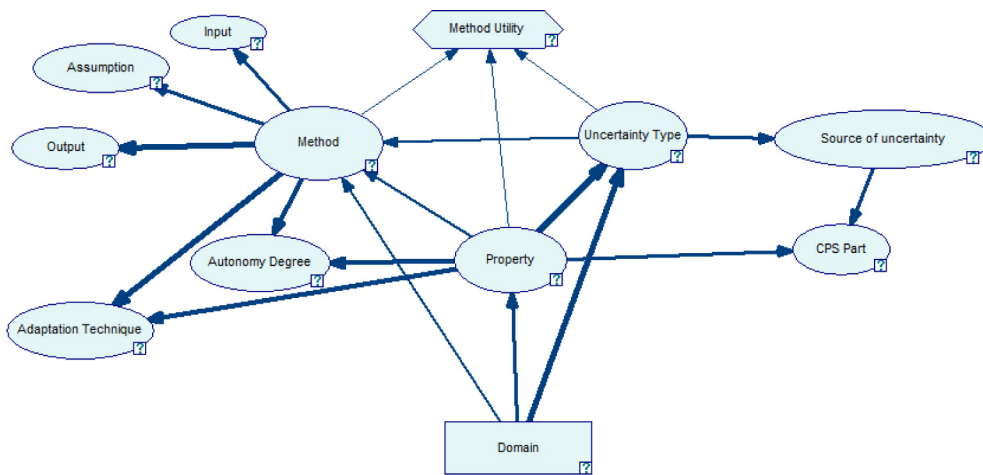


Fig. 8. Bayesian Network for Multi-Criteria Decision Making.

value should be represented separately in a new node with the values present/absent.

7.3. Taxonomy of methods for handling uncertainties

Building upon the answers of research questions, the observations that followed, and the influences strength in the network presentation, we conclude some questions that the developer should be able to answer during the selection of the methods to handle the uncertainties in any use case (Fig. 9). The questions are:

1. What are the sources of uncertainties handled by the method?
2. What are the types of uncertainties that the method handles?
3. What is the structure of the method(s) used?
4. What are the assumptions of the method?
5. What are the input types of the method?
6. What are the output types of the method?
7. Which adaptation technique the method is involved in?
8. In which autonomy degree is the method used?

We already discussed the sources of uncertainties, types of uncertainties, adaptation technique, and autonomy degree in detail throughout the study and represented their influence over

the selection of the methods (Fig. 8). Still, some uncertainties require a single or a combination of methods to handle them, as we conclude from the links provided during answering RQ3 and from the observations. Also, we have not discussed the inputs, outputs, and assumptions related to the selection of the methods. Thus, we discuss these aspects in detail (Appendix Table A.15 for details) as follows:

Structure. When choosing the methods to handle uncertainties, the developer should consider how the methods are used if there is more than one. For several methods, we define *static use* or *adaptive use*. For instance, a single method mitigates uncertainty by modifying related parameters such as lower the sampling time in case of minimizing missing information (T7-HYDROBIONETS), or by changing the structure such as changing communication interfaces in case of extensive delays (T12-IPAC). As for static use of methods, it is possible to predict occupancy depending on historical data by creating a Transition Matrix based on Markov Chains (MC), as mentioned above. Adaptive use of methods propose to have reinforcement learning that learns from outcomes of supervised learning methods for the home energy usage (T4-OrPHEuS). This type of method is usually responsible for updating the threshold that controls the adaptation decision-making process or having weights associated with the feedback loop.

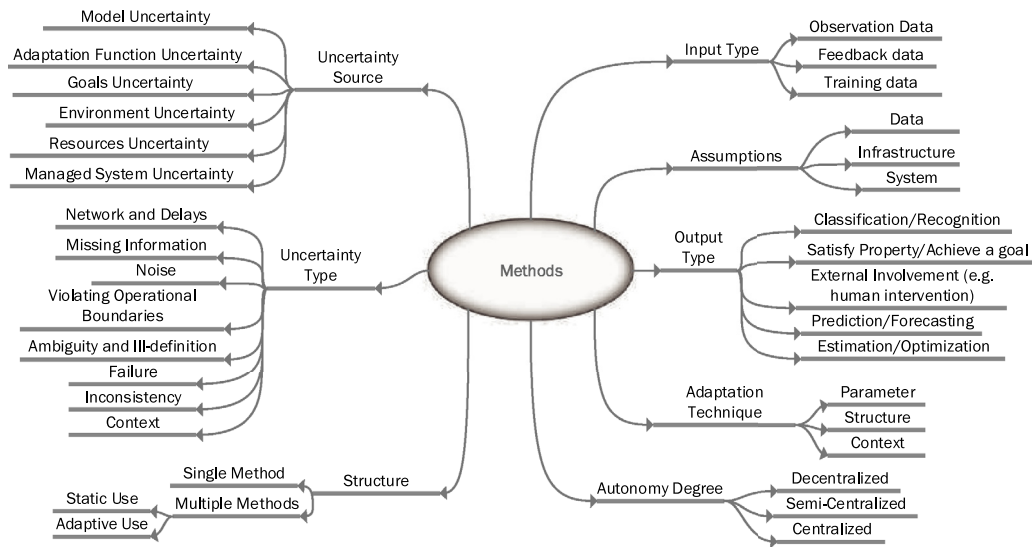


Fig. 9. Taxonomy of Methods to Handle Uncertainty.

Assumptions. Each method has its assumptions. Depending on the available data, needed output, and the scenario, the assumptions can be limiting. For instance, using Markov chains is not suitable when a trend over historical data is the target. The assumptions can be associated to *data*, *infrastructure*, or the *system* itself. For data, the assumptions include involved variables such as being (in-)dependent variables, their distribution, or associated weight (e.g., voting over many sources of values). Another example is the time series components (i.e., trend, seasonality, correlation, and multi-variant). Regarding the infrastructure, the assumptions are associated with resources such as the hardware configuration or limitation of communication (e.g., bandwidth or short-range communication). As for the system itself, examples of its assumptions are the system model and the probability associated with its states/transitions. Also, whether the system is (non-)linear or (non-)deterministic impacts the choice of suitable methods hugely.

Input type. The inputs to the methods can be *observation data* such as historical data, *training data* such as a (un-)labeled training data, and *feedback data* such as in control loops or involvement of human.

Output type. The output type determines the targeted results from using the method. Therefore, it is important to consider this part before going for further details. Here, we list the expected results from the methods covered by the study, which include: *estimation/optimization*, *classification/recognition*, *satisfy property/achieve goals*, *prediction/forecasting*, and *external involvement*. Learning is captured by having the input type as feedback data.

7.4. Employing the results to guide developers in CPS design

Designing a Cyber-Physical System (CPS) [24,25] requires multidisciplinary knowledge and multiple modeling paradigms for the development process stages such as gathering requirements, architecture design, and simulation. Looking into the CPS characteristics, the heterogeneity and the collaborative behavior are sources of many inherent uncertainties, which have varying impact on the system behavior. The impact of uncertainties on the system behavior can be explored through simulation to help the developers to quantify the impact of the uncertainties. In some cases, components in the system need to adapt to environmental changes and collaborate to satisfy the system requirements.

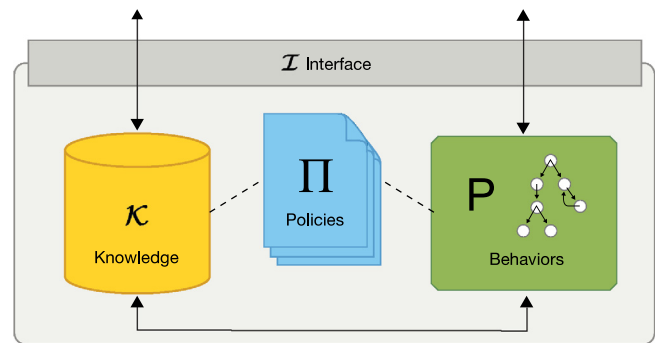


Fig. 10. Service Component.

Without proper management of the uncertainties, the decision-making process in the components is compromised and could even lead to faulty behavior. Thus, the developers need to ask the right questions to collect and trace the information needed for the system to make the right adaptation decisions at runtime. Consequently, the developers need guidance to properly support uncertainty awareness in the self-adaptive CPS components.

We present a guide that we have developed using mainly the information extracted from the systematic review. The guide links high-level and low-level concepts that need to be considered by developers. It describes a step-by-step process for designing uncertainty-aware CPS components. When following the guide, the developers will acquire the information needed to determine the impact of uncertainties in the adaptation decisions and select the appropriate methods to handle the uncertainties in the design of autonomous components. Finally, we demonstrate the use of the guide on an example from the projects.

7.4.1. Key concepts

When designing a CPS, the developer needs to identify the system requirements and the components that satisfy them. It is difficult to represent cooperation between components satisfying the requirements using a static architecture because the components may appear and disappear dynamically. Thus, the architecture should support dynamic component grouping. Given that the components need to adapt to the environmental changes,

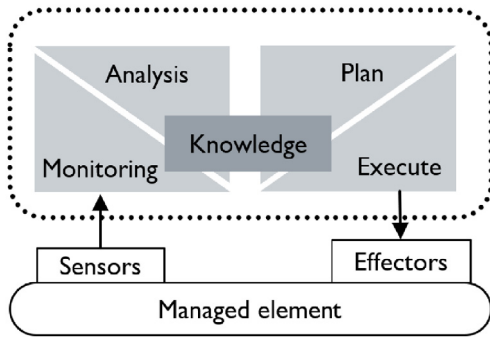


Fig. 11. MAPE-K Loop.

the developer is required to identify the impact of the uncertainties in the adaptation decisions. To address these concerns, the guide builds on the following key concepts:

- **Architecture.** A Cyber-Physical System is an emergent system that is distributed, decentralized, dynamic, self-adaptive, and scalable. It consists of components that dynamically form ensembles, which are defined by some common goal or other conditions. The components contain knowledge presented as interfaces and processes that execute the component behavior, while the ensembles are responsible for exchanging knowledge between the components. Each ensemble evaluates the constraints over the knowledge of the components to determine if the knowledge exchange is to happen. This structure provides dynamicity and context-awareness to the system because of the continuous knowledge update at runtime that causes updates in the evaluation of the ensembles' conditions. The concept of component ensembles is used in the EU projects selected in the study we conducted, namely the ASCENS project [26], in which the components are called Service Components (SCs) and the ensembles are called Service Component Ensembles (SCEs). The project proposes a formal language, Service Component Ensemble Language (SCEL) [27] (Fig. 10), which enables the definition of component ensembles using concepts such as Knowledge, Behaviors, Aggregations, and Policies.

- **Design Process.** To design any system, the developers need to identify the system requirements and build an architecture that satisfies them. During design, the developers should also consider the potentially faulty behaviors that can put the system in border situations (e.g. hazards [28]). Addressing border situations is an essential part of the system design to ensure that the system requirements will be satisfied. Adaptation is a method used to address border situations based on decisions evaluated over component knowledge. However, the knowledge holds uncertain information, which impacts the confidence in the decisions made and the resulting behavior.
- **Self-Adaptation Model.** The system must support adaptation to deal with environmental changes. The adaptation can happen at the level of the component providing the system with autonomic behavior. A well-known adaptation model, the MAPE-K loop, realizes adaptation phases over component knowledge as follows:

- **Monitoring.** For a component to be aware of its environment, it needs to sense it frequently. The frequency of monitoring varies depending on the domain and the situation in which the component is. The data obtained can be sanitized (e.g., filtered, aggregated) and stored in the Knowledge for further analysis.
- **Analysis.** The component analyzes the data stored in Knowledge to evaluate if adaptation in the component behavior is required to achieve the system goals.
- **Planning.** After the component evaluates the current situation and decides for adaptation, it plans to execute a set of actions to perform the adaptation to reach the system goals.
- **Execution.** In this phase, the component executes the planned actions that influence the physical world through effectors.
- **Knowledge** This part stores the component data used through all the model phases.

The MAPE-K loop has been used for representing self-adaptive behavior and interaction with physical elements, but it does not specify how to deal with uncertainties (Fig. 11) [29,30]. However, we support a view that we present later, based on the MAPE-K loop, showing the information related to uncertainties in CPS components and how to handle them.

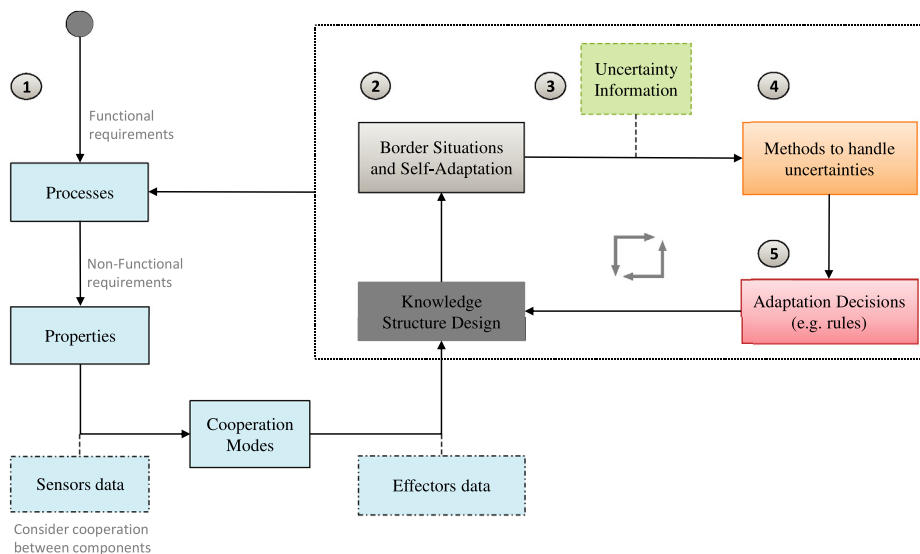


Fig. 12. An overview of the design guide for uncertainty-aware components in CPS.

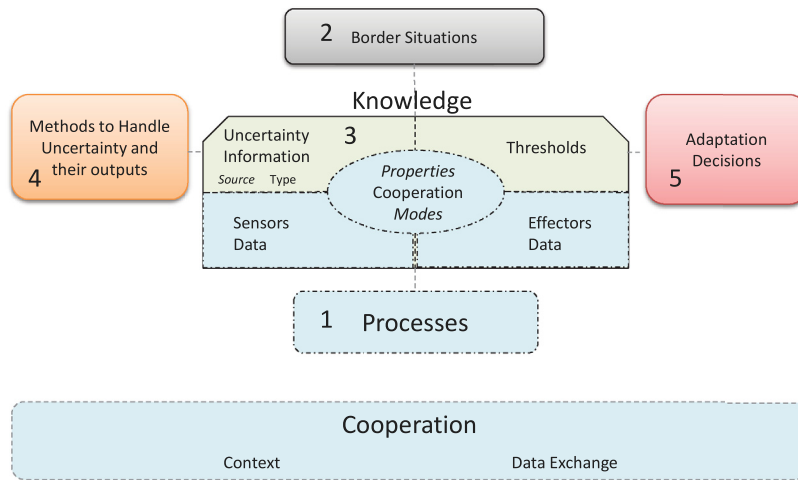


Fig. 13. A knowledge-oriented framework to design uncertainty-aware self-adaptive components in CPS. The framework includes five major steps including basic information of the physical component and its operations besides the included adaptation decisions and the uncertainties.

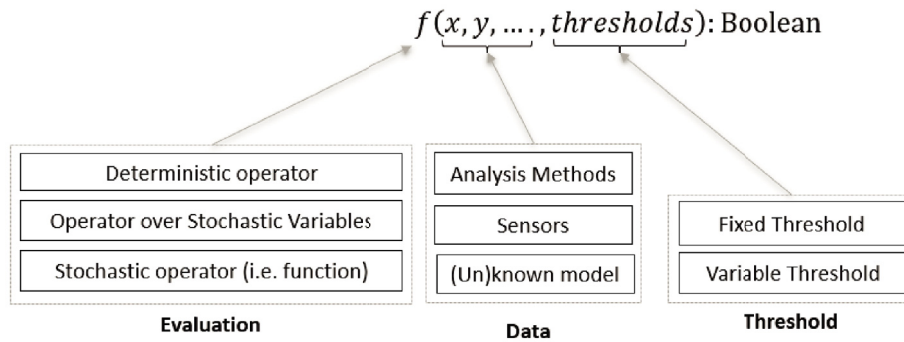


Fig. 14. The rule condition is defined in form $f(x, y, \dots, thresholds)$ and its evaluation is Boolean value.

- **Impact of Uncertainties on Adaptation Decisions.** We understand that CPS has many sources of uncertainties. However, we focus on the uncertainties that directly influence adaptation decisions, which means they either impact data passed to the decision-making processes or are related to models used to analyze the data in the decisions. Thus, we focus on the uncertainties associated with data stored in the Knowledge of MAPE-K loop and the selection of the methods to handle them. Based on this Knowledge-based view of MAPE-K, we support the developer with a systematic approach for incorporating handling of uncertainties into adaptation decisions. In our systematic approach for building an adaptation decision, we provide a lightweight notation that will enable the developers to see the impact of uncertainties in data on the adaptation decisions. The approach supports the selection of methods to handle uncertainties by using the study's findings.

7.4.2. Uncertainty-based framework for developers

In this section, we present the steps of developing autonomous components in CPS that consider uncertainties, and provide a framework based on the MAPE-K loop to help developers representing the needed knowledge regarding uncertainties in adaptation decisions. We suggest the following steps presented in Fig. 12 to design components inspired by aspects of Dependable Systems Design from Software Engineers [28], in which the system design contains identifying the hazards and capturing fault trees for analysis to ensure system dependability. The steps we propose are:

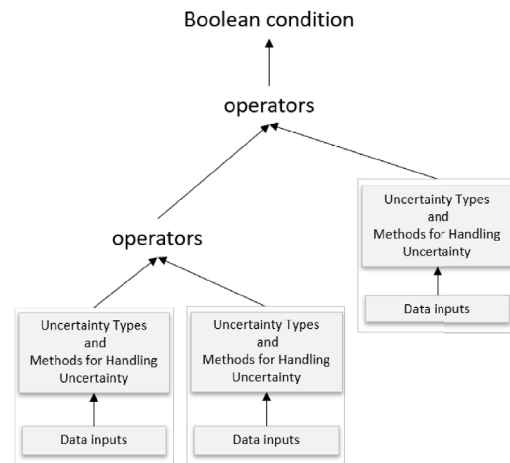


Fig. 15. The selected methods to handle uncertainty in adaptation decisions represented by lightweight notation.

1. Complete the steps of the guide in Fig. 12

- 1.1 Identify the key parts that constitute autonomous components in a CPS, i.e., processes, properties, inputs from local or cooperation sources, modes, and

output types. Then, fill the information on the knowledge framework (Fig. 13).

- 1.2 Identify the border situations and answer the self-adaptation questions [22].¹
- 1.3 Extract the sources and types of uncertainties, using the network presented in Fig. 8 as well as thresholds of the adaptation decisions.
- 1.4 Select the methods to handle uncertainties using the network presented in Fig. 8.
- 1.5 Determine the adaptation decisions (e.g., rules) (Figs. 14 and 15). Then, update and fill the information on the knowledge framework based on MAPE-K loop (Fig. 13).

2. Repeat the steps until the design is stabilized.

The decisions taken here are related to dealing with border situations, which requires mode-switching. We decided to represent mode switching as a rule in the following format: $f(x, y, \dots, thresholds) \rightarrow Mode$. The rule condition (Fig. 14) is defined in the form of $f(x, y, \dots, thresholds)$ and its evaluation is a Boolean value (i.e., could be combined using logical operators), where x, y , etc. are the input data, such as sensory information about an observed object/environment that could pass through data analysis; thresholds are the values to be used for comparison; the evaluation is an operator or a comparison function. If the condition evaluation returns true, it causes activating the mode associated with that condition.

More specifically, the information is gathered from local sensors and other components about the current state of the components themselves or the surrounding environment. The data could be analyzed later on, which requires a specific type of inputs depending on the method used, such as historical measurements, data points. To operate in specific context, it is important to understand the involved parts in the operating environment that may have known models such as vehicle movement or unknown models such as bacteria growth. Also, the involved hardware must be taken into account. For instance, each sensor has its precision and uncertainties (e.g., WiFi data has delays and radar data has missing information caused by road curves or hills). Usually, the shortcomings and precision is defined by the industry. The analysis is performed using a selected method to clean and perceive the collected information, where the results will be used in planning and decision making. Nevertheless, the methods themselves also can be error-prone. The evaluation of the rule condition should be taken into account while selecting the suitable operators: Whether the condition is deterministic (e.g., $>$, $<$), whether the variable is stochastic, and if the function is stochastic.

By identifying the types of uncertainties and identifying the outputs from handling them, we can select the methods to handle the uncertainties from Fig. 8, and order them by the sources and types of uncertainties in a tree form using a lightweight notation (see Fig. 15).

7.4.3. Demonstrate the usage of the results on an example

In this section, we illustrate the use of the developer guide when designing uncertainty-aware components. To this end, we apply the guide on a use case from the EU projects. We select the demonstrator in the OrPHEuS project as a representative example because it covers one of the most targeted domains,

with having different autonomy degrees and parameter adaptation techniques. The demonstrator also contains many uncertainties, including violating operational boundaries and use statistical methods and machine learning to handle them. In OrPHEuS, the demonstrator employs multiple systems to provide additional services or improve existing ones, presenting hybrid-energy management using heating, electricity, and fuel systems. We focus on the decisions related to energy distribution in houses that consumes and produces energy (i.e., prosumer).

The demonstrator is targeting the Energy domain, where a combination of heating, gas, and electricity is used to manage the energy response and demands. More specifically, the houses are prepared with photovoltaic panels, electrical and fuel boilers, and water-based storage. The photovoltaic panels depend on solar light to produce energy. The electrical boiler uses the surplus energy produced by the solar panels to provide hot water to the house and uses the rest thereof to heat the water-based storage (i.e., thermal storage). Additionally, each house that does not have night heaters can use central-night-electric-space heating offered at a discount price. These central space heaters are charged partially by the surplus energy from the houses during the day. The proposed structure allows for providing a layer of consumers that can act as producers (i.e., prosumers) by using their own power plants to provide energy to other consumers in the form of heating.

These devices are connected, and their control can be centralized, hybrid, or decentralized. For decentralized control of the energy demand and response, the house has a local surplus usage controller and another one to observe the local node voltage. For centralized control, the house uses central sources of heating, gas, and electricity. Also, the house preserves the surplus of locally generated energy from solar panels to heat the central space storage. To this point, there is no intervention of the uncertainties.

Uncertainties in the original demonstrator. Originally in the project demonstrator, the sources of uncertainties that should be taken into account are changes in resources, decentralization, and automatic learning. More specifically, the demand for energy changes depending on the context (U2, U8) (e.g., human usage of energy) that can lack knowledge (U2) about some variables, while the response (U4) of demand has time delays (U1) in the District Heating (DH) network and Combined Heat and Power (CHP).

In the demonstrator, weather forecasting is used to estimate and predict the energy production in the house. The energy consumption is based on the context (U2, U8) represented by consumer behavior, which forms a pattern of energy consumption for each home. The pattern or model of behavior is learned and predicted using Machine Learning and Neural Networks (HU4) that use the collected consumer profile to train. For violating operational boundaries, balancing the demand and response, taking into account the cost (U1, U4), the goal is to optimize and predict energy peaks and consumption and provide a suitable controlling algorithm that scales down production and reduces consumption. The response-demand balance is approached in the project by a combination of methods from the Statistics and Probability group (HU2), Re-Configuration group (HU3), and Machine Learning and Neural Network group (HU4).

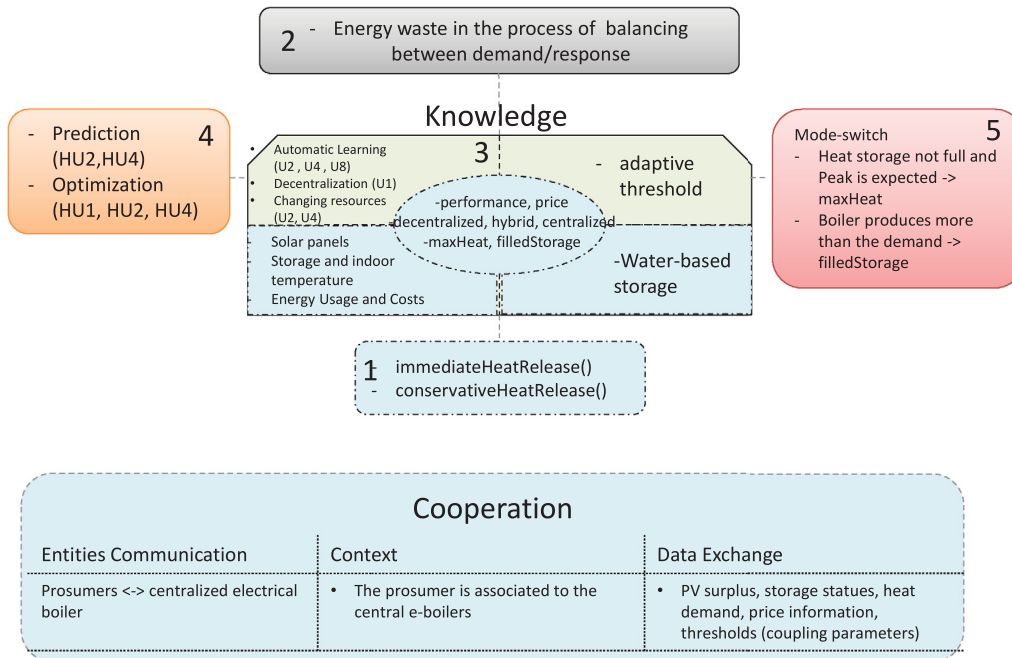
Using the guide. Assuming that the details about the demonstrator are already explained in the project (i.e., the business aspects), the developer can follow the steps in our guide and design a house as an autonomous component, and add the results to the framework presented in Fig. 16. The demonstrator specification presents the border situation to be energy waste in the process of balancing between demand and response. Based on that, the targeted properties are performance and price. The inputs are the data collected from solar panels, production, and energy

¹ When to adapt? Why do we have to adapt? Where do we have to implement a change? What kind of change is needed? Who has to perform the adaptation? How is the adaptation performed?

Table 13

Classification of the variables in collective sCPS design.

Domain	Unc. type	Method	CPS part	Adap. tech.	Autonomy degree
Smart Factory and Manufacturing	U1, U2, U5	HU1, HU2, HU3	All	All	Decentralized, centralized
Smart Homes and Buildings	U4	HU1, HU2, HU4	All	All	Decentralized, centralized
Smart Traffic and Transport	U4	HU2, HU4, HU6, HU7	All	All	Decentralized, centralized
Smart Grids and Energy	U1, U2, U4, U8	HU2, HU3, HU4, HU6	All	All	All
Robotics	–	–	–	–	–
Crisis and Emergency	–	–	–	–	–
Clouds	U4	HU2, HU7	All	All	Decentralized, centralized

**Fig. 16.** OrPHEuS Framework: The prosumer as a component.

consumption, including electricity usage and heating. The component aims at optimizing and predicting the consumption and production of energy by managing the heating system. Starting from this stage, the developer can use the network to try different options and explore the targeted uncertainties and the methods to handle them.

The types of uncertainties inside this domain are the same as those in the project demonstrator; however, there is an additional method used by another demonstrator, which is Human in the loop (HU6), to handle violation of the operational boundaries (i.e., GreenCom where human reacts manually to the energy consumption). The developer can choose other domains and add evidence to properties, such as design choice (i.e., performance and price). The results in Table 13 show Ambiguity and Ill-Definition (U5) as an additional targeted uncertainty type and methods Data Filtering and Estimation (HU1), Human-in-the-loop (HU6), and Declarative Programming (HU7) as the used methods. The probability is displayed in front of each option in the network, which helps the developers decide whether to consider or ignore some aspects. For instance, more than 49% of U4 and U8 are addressed in the Smart Grids and Energy domain, while U1 and U2 are around, making the U4 and U8 essential to consider in the use case. The developer can consider this additional information in his/her design, e.g., by having ambiguity in identifying the patterns in occupancy of home and the energy usage caused by huge

space of variables. Of course, the developer can try adding other evidence to other aspects in the network and repeat the process such as adaptation technique or even focus on one uncertainty type to explore the handling methods.

After having the framework ready, the developer can approach the adaptation decisions. Taking Fig. 17 as an example, we suggest using Declarative Programming (HU7) (i.e., ensembles in this case) to form a dynamic knowledge exchange for exporting the heat. Assuming that the energy usage data is simplified to be seasonal, correlated, and having a trend, energy consumption can be estimated and predicted using Statistical methods (HU2), such as operators based on ARIMA models (the details are out of the scope of this article). Regarding violating operational boundaries, we propose using a learning approach, based on Statistical methods (HU2), that uses operators of correlation coefficients between the inputs and the outputs to choose the most suitable control strategy (i.e., centralized/decentralized control) for balancing the demand/response.

As we can see, the guide provides a practical framework to represent the relevant information to design self-adaptive cooperative CPS components. The network helps the developer try different options and consider issues and solutions from other domains, which can be time-consuming and require personal experience.

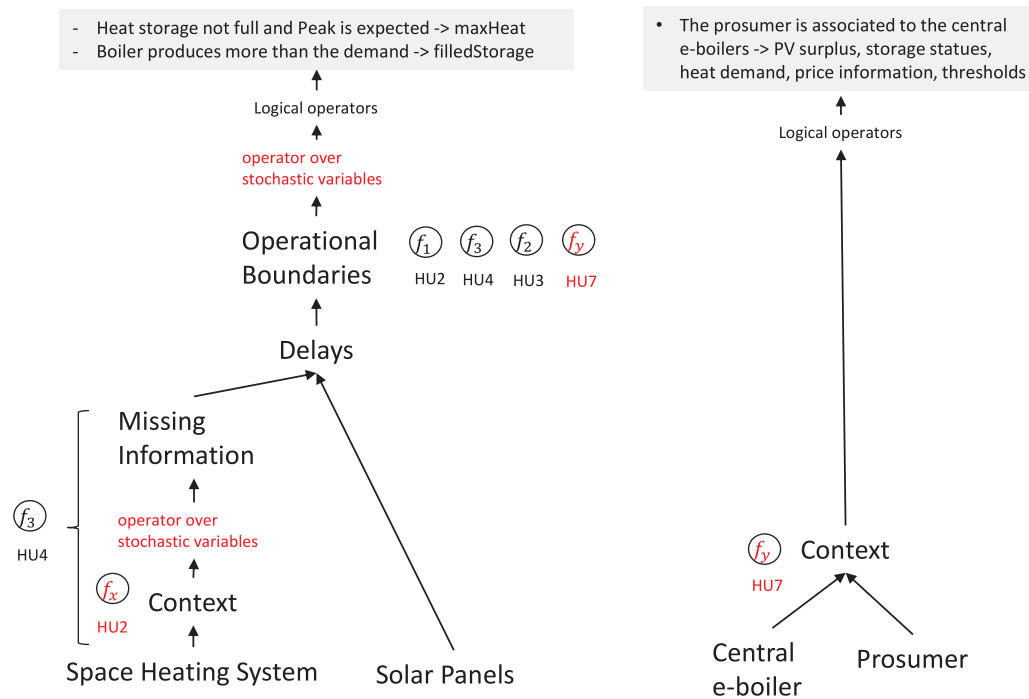


Fig. 17. Impact of uncertainties on adaptation decisions of a prosumer house. Red: our solution with the operators; black: project solution.

Table A.14

Self-adaptation information.

Project	Domains	Properties	When?	Why?	Where?	What?	Who?	How?
VICINITY	D2, D3, D4	P2, P3	Proactive, Reactive	User, resources	E, M	Parameter, Structure	Node, Cloud	Decentralized, Centralized, Rules
symbIoT	D2, D3	P1, P2, P4	Proactive, Reactive	Context, resources	E, M	Parameter, Structure	Node	Decentralized, Rules
CADDY	D5	P1, P4, P5	Reactive	User, resources	E	Parameter	Node	Decentralized
OrPHEuS	D4	P2, P3	Proactive	Resources	E	Parameter, Structure	Cloud, Edge	Decentralized semi-centralized and centralized, rules
ClouT	D7	P4	Reactive	Resources	E, M	Parameter	CPaaS Layer	Decentralized, rules
RECONFIG	D5	P4, P5, P6	Reactive	Resources	E	Parameter	Node	Decentralized
HYDROBI-ONETS	D1	P3, P4, P6	Proactive, Reactive	Context, resources	E	Parameter, Context	Gateways, Micro-servers, Nodes	Centralized, Cluster, Distributed
INERTIA	D4	P2	Proactive	Resources, user	E, M	Parameters, Context	Hubs	Decentralized, Rule-based decisions
NiFTi	D6	P4, P5	Proactive, Reactive	Context, user	E	Parameter	Node	Decentralized
GreenCom	D4	P2, P3, P4	Proactive	Context	E	Structure	Microgrid	Decentralized, rules and policies
ASCENS	D3, D6, D7	P2, P3, P4, P5	Proactive, Reactive	Context, resources	E	Parameter, Structure	Node	Decentralized
IPAC	D1, D3, D6	P1, P2, P5, P6	Reactive	Context, resources	E, M	Structure	Node	Decentralized, rules and policies

Domains - D1: Smart Factory and Manufacturing, D2: Smart Homes and Buildings, D3: Smart Traffic and Transport, D4: Smart Grids and Energy, D5: Robotics, D6: Crisis and Emergency, D7: Clouds;

Properties - P1: Safety, P2: Performance, P3: Price, P4: Resilience, P5: Cooperative, P6: Connectivity;

Why? resources: Change in the technical resources; context: Change in the context; user: Changes caused by user(s);

Where? E: Ensemble of Applications, M: Middleware.

Table A.15
Answering method selection questions.

Project	Uncertainty source	Uncertainty type	Structure	Assumptions	Inputs	Outputs
VICINITY	Statistics and Probability, Machine learning and Neural Network, Human-in-the-loop	U4, U8	Multiple-Methods (static use)	Data	observation data	Optimization, Prediction, External Involvement, Recognition
symbIoT	Data Filtering and Estimation, Machine learning and Neural Network	U2, U3, U4, U6	Multiple-Methods (static use)	Infrastructure, System	observation data	Estimation, Prediction, External Involvement
CADDY	Data Filtering and Estimation, Re-Configuration, Machine Learning and Neural Network, Human-in-the-loop, Declarative Programming, Verification	U3, U5, U7	Multiple-Methods (Adaptive use)	System, Data	observation data, training data, feedback data	Estimation, Optimization, Classification, Recognition, Satisfy Property/Achieve goals
OrPHEuS	Statistics and Probability, Machine learning and Neural Network, Re-Configuration	U1, U2, U4, U8	Multiple-Methods (Adaptive use)	Data, Infrastructure	observation data, training data, feedback data	Forecast /Prediction, Optimization
ClouT	Statistics and Probability	U5, U6	Multiple-Methods (Adaptive use)	Data, System	observation data	Classification
RECONFIG	Declarative Programming, Statistics and Probability, Machine Learning and Neural Network, Data Filtering and Estimation	U1, U3, U5, U6	Multiple-Methods (static use)	Data, System	training data, observation data	Satisfy Property/Reach a goal, Classification, Estimation, Recognition
HYDRO-BIONETS	Machine learning and Neural Network, Data Filtering and Estimation, Re-Configuration, Statistics and Probability	U1, U2, U3, U5, U6	Multiple-Methods (Adaptive use)	Data, System, Infrastructure	training data, observation data	Prediction, Estimation, Optimization
INERTIA	Statistic and Probability	U4, U7, U8	Multiple-Methods (static use)	Data, System	observation data	Recognition, Optimization, Prediction
NiFTi	Statistics and Probability, Data Filtering and Estimation, Machine Learning and Neural Network, Human-in-the-loop	U1, U2, U3, U7, U8	Multiple-Methods (static use)	Data, System	training data, observation data, feedback data	Classification, Estimation, Optimization, External Involvement
GreenCom	Statistics and Probability, Machine learning and Neural Network, Human-in-the-loop	U4, U8	Multiple-Methods (Adaptive use)	Data	training data, observation data	Prediction
ASCENS	Statistics and Probability, Declarative Programming, Verification	U4, U6, U8	Multiple-Methods (static use)	Data, System	observation data	Satisfy Property/Achieve goals, Prediction, Optimization
IPAC	Statistics and Probability, Re-configuration, Data filtering and Estimation, Declarative Programming	U1, U2, U3, U7	Multiple-Methods (static use)	Data, Infrastructure	observation data	Optimization, Estimation

8. Threats to validity

In this section, we discuss the threats to the validity of our study, which we divide as follows:

Construct Validity. The first threat is the selection of the primary studies (projects). The SLR guidelines suggest that the primary studies be peer-reviewed papers. However, peer-reviewed papers guarantee neither the existence of real industry-relevant demonstrators nor do they provide a detailed description of relevant use cases that our study requires. According to [16], the types of research questions in the study determine what the best selection for primary studies is. We, therefore, chose projects instead of peer-reviewed papers. More specifically, we targeted EU projects in ICT calls because projects accepted in these calls are required to deliver real demonstrators (i.e., TRL scale), and because industrial leadership makes them more likely to be industrially relevant. It is also important to keep in mind that each project has scientific results in addition to a running demonstrator. Using the deliverables, we access both the results of their publications and the overall coherent description of the use cases. Also, we ensure to count the relevant use case just once even though it is mentioned in several publications. Although the study is limited to EU projects, some calls target collaboration

with international parties, which we also considered. The well-known structure of the Horizon2020 and FP7 projects allows us to illustrate how the projects could be used in systematic reviews.

Another threat is related to our selection of project calls. Other calls than ICT might also deliver real demonstrators with industrial orientation. For instance, projects accepted under Innovation in SMEs, Future and Emerging Technologies (FET), or Marie Skłodowska-Curie Actions calls. To mitigate this threat, we did a pre-selection targeting CPS projects in all the calls. The results showed that ICT calls had the highest number of relevant projects.

Concerning the selection criteria, a threat is related to the choice of search keywords. Each criterion could have been based on another set of keywords on which the inclusion and exclusion of the projects would depend. To minimize this threat, we started with a much larger selection of keywords and applied it to three calls only. Nevertheless, the results did not show any significant impact on the final selection of related projects. Therefore, we dropped keywords with a too-broad meaning across different domains, which has not impacted the final selection.

Internal Validity. Another threat is related to the domain selection of the use case. There are intersections between the domains, and the state-of-the-art lacks a clear definition of these domains

and the boundaries between them. This issue could have led to different results. For this reason, we defined a set of domains that are the most targeted in CPSs and put a supporting example for each one. For instance, interoperability between different IoT platforms using a cloud, each one targeting a different domain (i.e., Smart Traffic and Transport, Smart Homes and Buildings), is under the Cloud domain in the ClouT project. Another example is robotics, as in the case of RECONFIG. We chose this domain when the robots are not used in another specific application, such as a part of a rescue team during a crisis.

Conducting Validity. The last threat we would like to mention is handling the challenges. The problem here lies in the level of abstraction that a project and its use case description use. A high-level description of a use case is not sufficient to determine all the challenges that the use case addresses. To tackle this issue, we defined quality assessment that helps us in rectifying the conducting.

External Validity. We cannot claim that our approach is general, because it is based on a limited set of solutions from selected projects and needs further validation. Nevertheless, it captures the basic concepts that are necessary to design autonomous components with considering uncertainties. Also, the study shows the possible interesting future research directions. For instance, a closer look at the inputs and assumptions of the methods could be extended in a more extensive and detailed study.

9. Conclusion

Designing smart cyber-physical systems capable of dealing with uncertainties in the data representing the system's perception of the environment can be a daunting task. Even identifying the existence and the type of uncertainties in the data and choosing a suitable method for mitigating the effects of uncertainty requires a considerable breadth of knowledge. To simplify this aspect of CPS design, we aim to develop a methodology that helps to approach the problem of identifying particular types of uncertainties and the corresponding methods for mitigating their effects in a systematic fashion.

To this end, we present a systematic review of EU-funded industry-led research projects dealing with collective behavior of smart cyber-physical systems, which identifies the links between different high-level aspects and the methods used to handle uncertainties in such systems. Our study is atypical in that we apply the methodology for conducting a systematic literature review to descriptions and outputs of research projects (demonstrators and reports) instead of scientific papers. This choice was made intentionally to draw on the challenges and requirements considered relevant by industrial partners who are more likely to build actual systems, and who had to create a working demonstrator as a deliverable.

The results of our study provide an overview of uncertainties across various domains and challenges, exploring the dependency on the domain-specific terminology, context, and requirements. This makes it easier to relate them to CPS designs in different domains, dealing with different challenges using similar models of uncertainties and uncertainty-handling methods. We also presented a design guide for developers that stemmed from the existing information in the projects. This guide provides steps to build the self-adaption rules in CPS components taking into account uncertainties.

During the study, we aimed at helping the developers in their design process of collective sCPS components taking uncertainties into account. However, the work would benefit from a further investigation of the other possible factors related to uncertainties and the selection of methods. More specifically, the factors that we see interesting as a next step are relations between the uncertainties themselves, relations between the methods themselves,

and the computation requirements. The relations between the uncertainties can provide a deeper insight into the design of similar demonstrators. In contrast, the relations between the methods provide knowledge about the possible structures of methods used to handle the uncertainties and capture the structure and computational requirements for each option. More elaboration over representing the notion of representing the uncertainties and the methods in self-adaptation decisions are needed and validation for the guide through a controlled quantitative study.

CRedit authorship contribution statement

Rima Al-Ali: Idea, Methodology, Majority of the paper text. **Lubomír Bulej:** Proposed methodology, Writing parts of the paper (Introduction, conclusion). **Jan Kofroň:** Proposed methodology, Writing parts of the paper, Figures (methodology, proposed paper contribution). **Tomáš Bureš:** Discussing the ideas, Proposing the paper direction.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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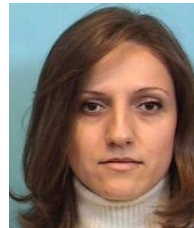
Appendix. Supplementary tables

See [Tables A.14](#) and [A.15](#).

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